

PRIMARY 4 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

FIRST TERM

Week	Topic	Content
1	Changes in Nature	Observation of changes in nature - day and night, seasons, weather patterns; Recording daily weather conditions; Understanding natural cycles
2	Developmental Changes in Plants	Life cycle of plants; Germination process; Parts of a flowering plant; Growth stages from seed to mature plant
3	Growth and Developmental Changes in Animals	Life cycles of common animals; Stages of growth in humans; Changes from birth to adulthood; Comparing growth in different animals
4	Exploring Water in Our Environment	Sources of water; Uses of water; Water bodies around us; Importance of clean water; Simple water conservation methods
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Human Body: The Mouth	Parts of the mouth; Functions of different parts; The tongue and taste; Proper oral hygiene; Speaking and eating functions
8	Human Body: The Teeth	Types of teeth and their functions; Permanent and milk teeth; Dental care and hygiene; Common dental problems; Healthy eating for strong teeth
9	Non-Living Things: Rocks	Characteristics of rocks; Types of rocks; Uses of rocks in daily life; Rock formations in the environment
10	Properties of Living and Non-Living Things	Differences between living and non-living things; Movement, nutrition, respiration in living things; Comparing characteristics
11	REVISION	Review of all topics covered; Practice exercises; Question

		and answer sessions
12	EXAMINATION	End of term assessment
13	CLOSING	Report card distribution; Vacation assignment



PRIMARY 4 BASIC SCIENCE AND TECHNOLOGY LESSON NOTES

FIRST TERM

WEEK 1: CHANGES IN NATURE

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Mention examples of changes in nature (day, night, seasons).
2. Describe the differences between day and night.
3. Identify weather conditions (sunny, rainy, cloudy, windy).
4. State the importance of observing changes in nature.

CONTENT

Changes in Nature

Nature is everything around us that God made, not things people built. Think of the sky, trees, rivers, mountains, animals, and the sun. These things do not stay the same forever; they change all the time. We can see these changes if we look carefully.

- **Day and Night:** This is a change that happens every single day. It is caused by our Earth turning around like a spinning ball.
 - **During the Day:** Our part of the Earth faces the Sun. The sky is bright and blue. We can see clearly. The Sun gives us light and warmth. People are awake, working, going to school, and playing.
 - **During the Night:** Our part of the Earth turns away from the Sun. The sky becomes dark. We see the moon and stars. The moon is not a source of light; it reflects the Sun's light like a mirror. People and animals usually sleep at night.



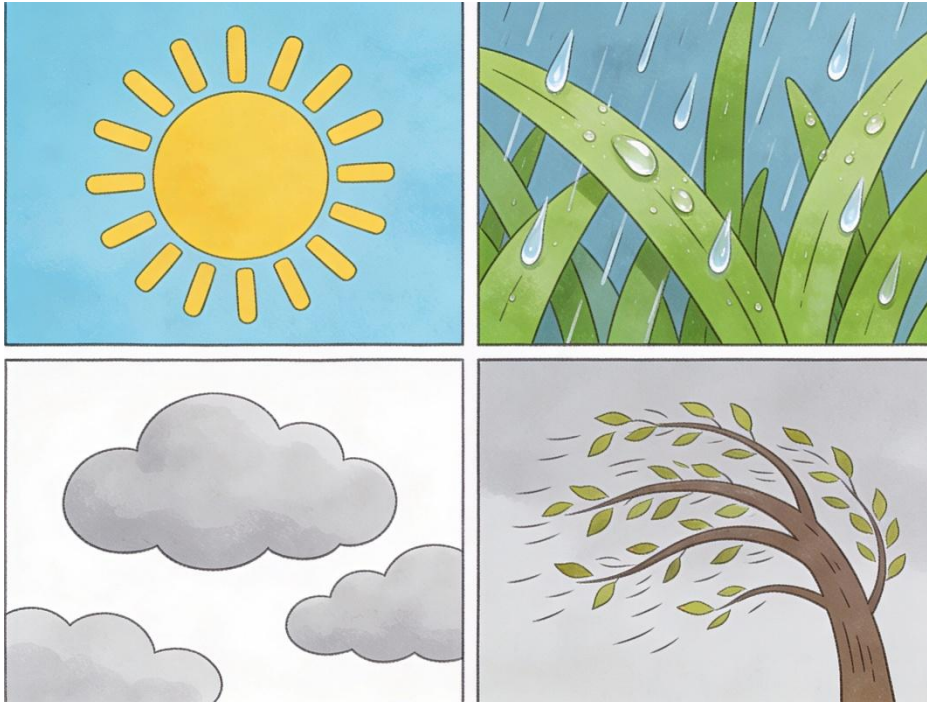
- **Seasons:** These are big changes in weather that last for months. In Nigeria, we have two main seasons:
 - **Dry Season (November - March):** The weather is hot and the air can be dusty, especially during Harmattan. There is little or no rain. The land and grass can become brown and dry. We wear light clothes and use moisturizer for our skin.
 - **Rainy Season (April - October):** It rains often, sometimes with thunder and lightning. There is plenty of water. Plants grow very well, and farms are green and full of crops. We use umbrellas, raincoats, and boots.

Daily Weather Conditions

Weather is how the air around us feels at a particular time. It can change from morning to afternoon! We describe it by looking at the sky and feeling the air.

- **Sunny:** The sky is clear, and the Sun is shining brightly. It feels warm or hot. Shadows are very dark and short.
- **Rainy:** Clouds are dark, and water falls from the sky as rain. We hear pitter-patter sounds on the roof. The ground becomes wet.
- **Cloudy:** The sky is covered with grey or white clouds. The Sun is hidden, so it is not very bright or hot. It might feel cool.

- **Windy:** We can feel the air moving. It blows leaves off trees, moves our clothes on the line, and can make dust fly. We hear a "whoosh" sound.



Importance of Observing Changes

Why should we pay attention to these changes?

- **For Farmers:** They watch the seasons to know the best time to plant seeds (start of rainy season) and the best time to harvest crops (during dry season).
- **For Us (Choosing Clothes):** We check the weather before dressing. On a sunny day, we wear a hat and light clothes. On a rainy day, we carry an umbrella.
- **For Planning:** It helps us plan our day. If it is raining, we know we cannot play football outside. If it is a sunny Saturday, it is a good day to help wash and dry clothes outside.

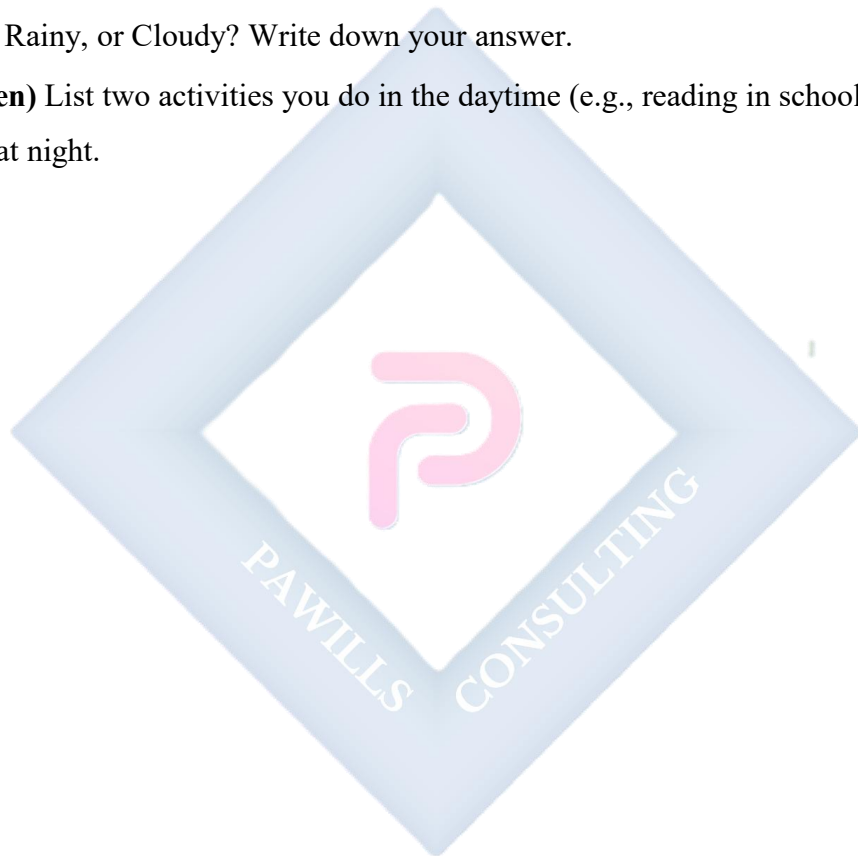
EVALUATION

1. Name two changes in nature that happen every day.
2. What makes the sky bright during the day?
3. Which season in Nigeria is good for farmers to plant their seeds? Why?

4. If you see dark clouds and feel water dropping on your head, what is the weather condition?
5. How does knowing the weather help your mother at home?

ASSIGNMENT

1. **(Drawing)** Draw two pictures side-by-side: one showing what you see in the sky during the day, and one showing what you see at night.
2. **(Practical/Observation)** Look outside your window tomorrow morning. Is the weather Sunny, Rainy, or Cloudy? Write down your answer.
3. **(Written)** List two activities you do in the daytime (e.g., reading in school) that you do not do at night.



WEEK 2: DEVELOPMENTAL CHANGES IN PLANTS

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define germination.
2. List the stages of plant growth from seed to mature plant.
3. Identify the main parts of a flowering plant.
4. Draw a simple plant showing roots, stem, and leaves.

CONTENT

Life Cycle of Plants

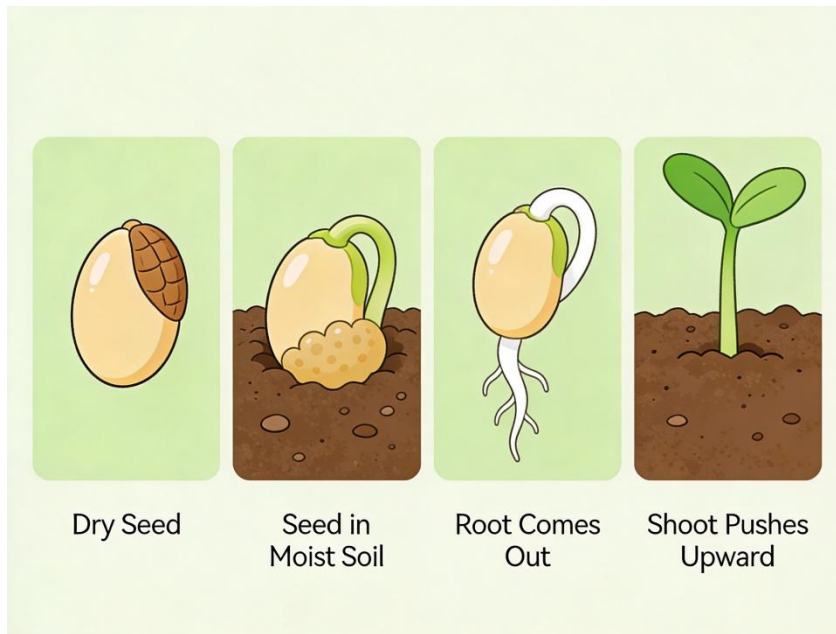
Plants are living things. Just like a baby grows into an adult, a plant also grows and changes. Most plants start their life as a tiny seed.

Germination Process

Germination is the wonderful process when a sleepy seed wakes up and starts to grow into a young plant. It is the beginning of the plant's life.

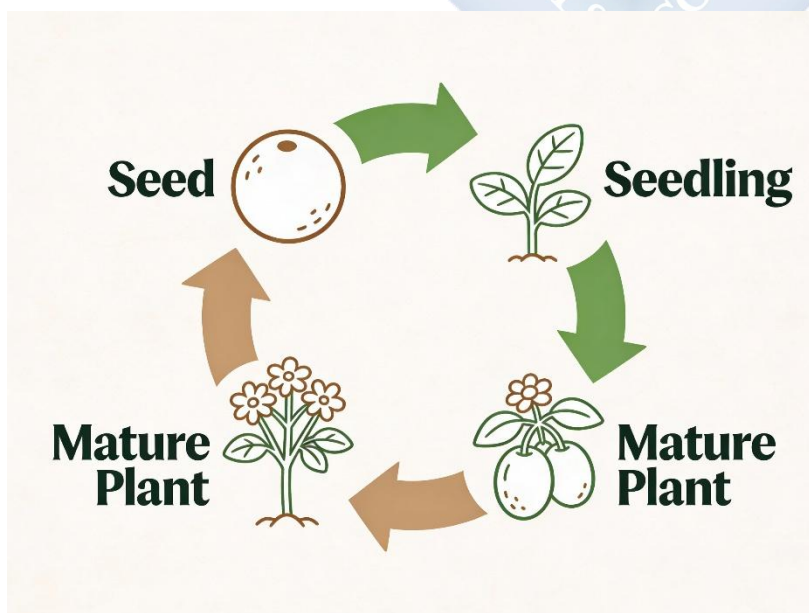
For a seed to germinate and wake up, it needs four important things:

- **Water:** To soften the hard seed coat.
- **Air (Oxygen):** For the baby plant to breathe.
- **Warmth (from Sunlight or soil):** It must not be too cold.
- **Good Soil:** To hold the seed and give it some food.



Growth Stages of a Plant

1. **Seed:** The beginning. It is small, dry, and hard. Inside is a tiny baby plant and its food.
2. **Seedling:** The young plant. First, a tiny root comes out to drink water. Then, a small shoot with tiny leaves pushes up towards the light. It is very delicate.
3. **Mature Plant:** A full-grown, strong plant. It has many roots, a thick stem, lots of leaves, and can produce beautiful flowers and fruits/seeds. The seeds can then start a new life cycle.



Parts of a Flowering Plant and Their Jobs

- **Roots:** Usually under the soil. They hold the plant firmly in the ground like an anchor. They also suck up water and minerals from the soil.
- **Stem:** Stands up tall. It holds the leaves and flowers. It acts like a pipeline, carrying water from roots to leaves and food from leaves to other parts.
- **Leaves:** They are the plant's kitchen. They use sunlight, air, and water to cook food for the plant (this is called photosynthesis).
- **Flowers:** They are often colourful and smell nice. Their main job is to make fruits and new seeds for more plants.

EVALUATION

1. What does 'germination' mean?
2. If you want to plant a seed, why should you put it in a place with sunlight?
3. What is the name for a young, small plant that has just started growing?
4. Which part of the plant is usually hidden under the soil?
5. Why are leaves very important for a plant?

ASSIGNMENT

1. **(Drawing)** Draw a full flowering plant. Label these four parts: Root, Stem, Leaf, Flower.
2. **(Practical Experiment)** Plant a bean seed in a small tin/cup with wet soil. Place it where it can get sunlight. Water it a little every day for 3 days and write down what you see each day.
3. **(Written)** Write one difference between a dry seed and a seedling.

WEEK 3: GROWTH AND DEVELOPMENTAL CHANGES IN ANIMALS

OBJECTIVES

By the end of the lesson, pupils should be able to:

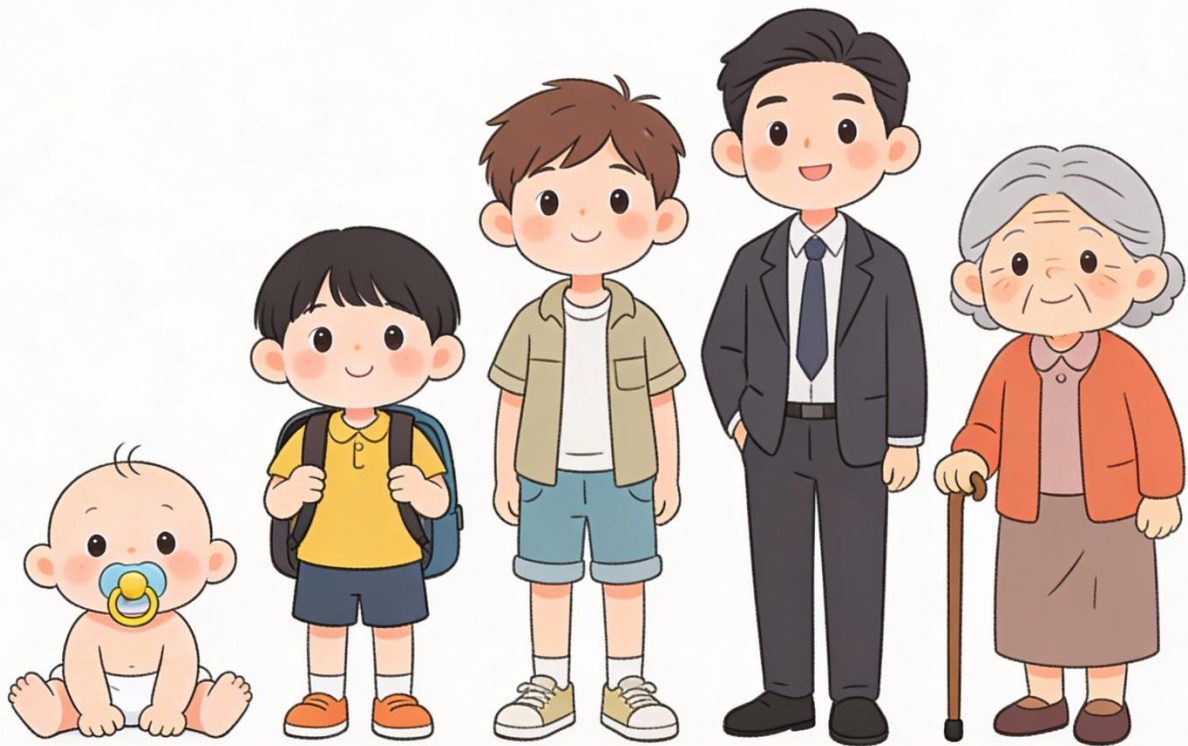
1. Describe the stages of growth in humans (Baby to Adult).
2. Mention the life cycle of a common animal (e.g., Insect or Chicken).
3. Compare the young of different animals to their adults.
4. State one difference between a baby and an adult human.

CONTENT

Growth in Humans

From the time we are born, we keep growing and changing until we are old. This is called human development.

- **Baby (Infant):** A newborn. Cannot walk or talk. Cries to communicate. Needs milk, sleeps a lot, and depends completely on parents for everything.
- **Child (Primary School Age):** Can walk, run, talk clearly, and learn in school. Loses milk teeth and grows new ones. Loves to play.
- **Adolescent (Teenager):** The body changes quickly to become an adult. Boys' voices become deeper, girls' bodies change. They become taller and stronger.
- **Adult:** A fully grown man or woman. Can work, get married, and have babies of their own. They are responsible.
- **Aged (Old Person):** Hair turns grey or white. Skin has wrinkles. The body is not as strong as before, but they are wise with experience.



Baby

Child

Teenager

Adult

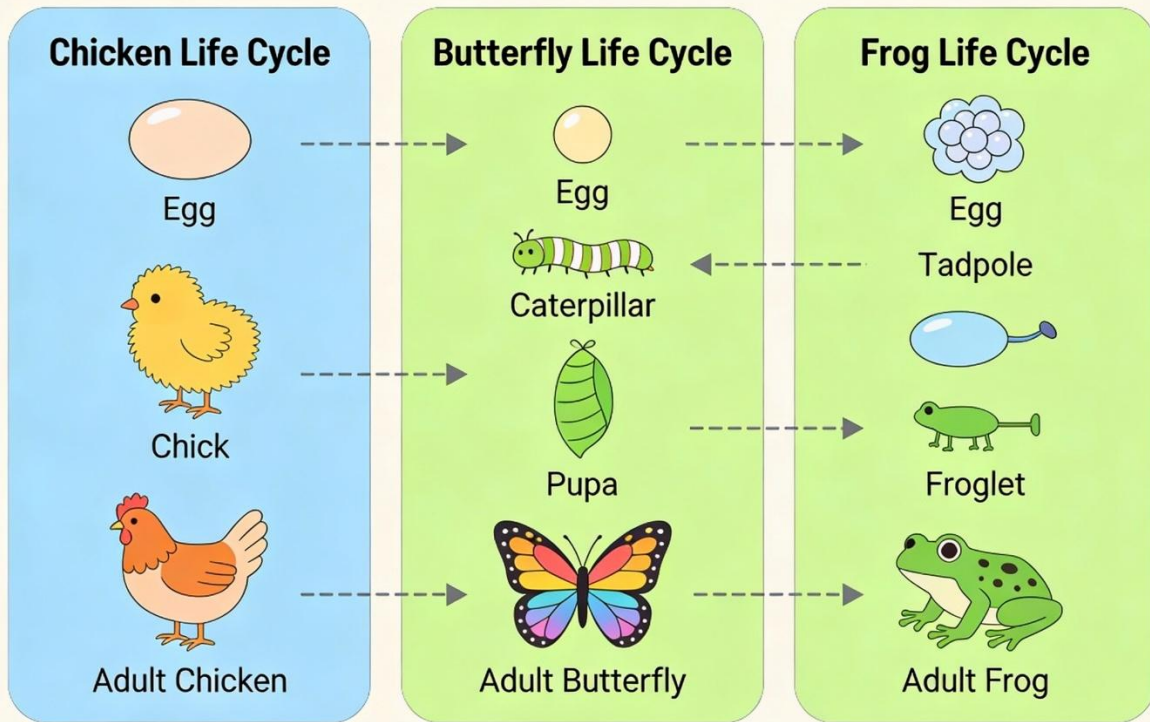
Old Person

Life Cycles of Animals

Animals also grow through different stages. This is their life cycle.

- **Chicken:** Egg --> Chick (fluffy, small) --> Hen (female) or Cock (male).
- **Butterfly:** Egg (on a leaf) --> Larva (Caterpillar - eats leaves) --> Pupa (Chrysalis - rests in a cocoon) --> Adult Butterfly (flies with wings).
- **Frog:** Egg (in water) --> Tadpole (lives in water, has a tail, no legs) --> Adult Frog (lives on land and water, has legs, no tail).

Animal Life Cycles Chart



Comparing Young Animals and Adults

- **Look-Alike Young:** Some young animals look like small versions of their parents.
 - Example: A puppy looks like a small dog. A kitten looks like a small cat. A human baby has features like its parents.
- **Different-Looking Young:** Some young animals look completely different from their parents.
 - Example: A tadpole looks like a little fish, but an adult frog has four legs and no tail. A caterpillar looks nothing like a butterfly.

EVALUATION

1. List three stages in human growth, starting from birth.
2. What comes out of a butterfly's egg? Is it a small butterfly?
3. Why does a baby need more care than a child?

4. Name one thing an adult can do that a baby cannot do.
5. Does a chick look more like an egg or like a hen?

ASSIGNMENT

1. **(Drawing)** Draw the life cycle of a chicken in three simple pictures: Egg, Chick, Hen.
2. **(Written)** Write the names of two animals and what their young ones are called (e.g., Goat - Kid, Cat - Kitten).
3. **(Oral/Reflection)** Ask your parent or guardian: "At which stage of human growth are you?" (Baby, Child, Adolescent, Adult, Aged).



WEEK 4: EXPLORING WATER IN OUR ENVIRONMENT

OBJECTIVES

By the end of the lesson, pupils should be able to:

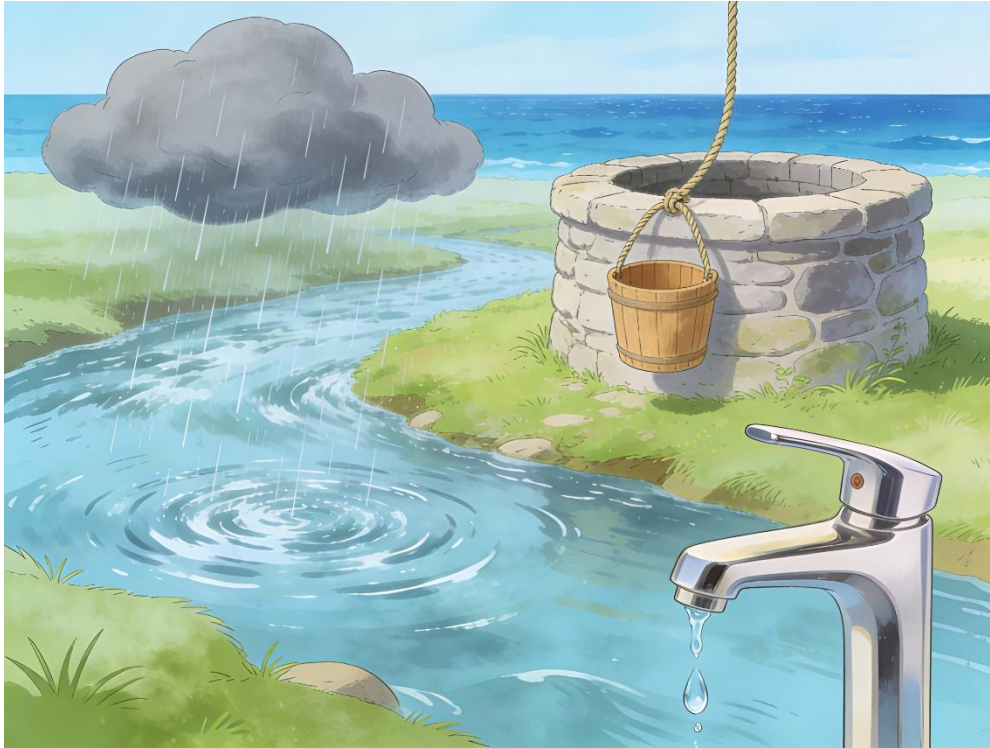
1. List three sources of water.
2. Mention four uses of water in the home and school.
3. Identify water bodies in the environment.
4. State two simple ways to keep water clean.

CONTENT

Sources of Water

Where does the water we use come from?

- **Rain:** Water that falls from clouds in the sky. We can collect it with bowls or tanks.
- **Rivers and Streams:** Long, flowing bodies of fresh water. People fetch water from here in some villages.
- **Wells:** Deep holes dug into the ground to reach underground water. A bucket is used to draw water.
- **Boreholes/Taps:** A machine drills very deep to pump water up. The water is stored in tanks and comes to our taps at home.
- **Lakes and Ponds:** Still water collected in a large area.
- **Oceans and Seas:** Very, very large bodies of **salty water**. We cannot drink this water.



Uses of Water

Water is essential for life. We use it for many things every day:

- **Drinking and Cooking:** We need clean water to drink, cook rice, stew, and soup.
- **Cleaning:** We use water to wash our bodies (bathing), wash our clothes, plates, and our houses.
- **Farming (Agriculture):** Water is used to water crops like vegetables, maize, and yam so they can grow (irrigation).
- **Transportation:** Big boats and ships carry people and goods across rivers, lakes, and oceans.
- **Recreation:** We swim and play in water for fun.

Clean Water and Conservation

Dirty water has germs that can cause sicknesses like cholera and typhoid.

- **Keeping Water Clean:** Always cover stored water in pots or tanks. Do not throw rubbish or bath inside rivers where people fetch drinking water.

- **Water Conservation:** This means using water wisely and not wasting it.
 - Turn off the tap tightly after use.
 - Fix a leaking tap or bucket immediately.
 - Use only the water you need for washing.

EVALUATION

1. Name two sources of water that you can see in our community.
2. How do we use water in the kitchen?
3. Why is it dangerous to drink water from a dirty stream?
4. What is the difference between a well and a borehole?
5. What should you do if you see a tap dripping water?

ASSIGNMENT

1. **(Drawing)** Draw a bucket filled with water and a cup beside it.
2. **(Written)** List three different ways you used water from morning until now.
3. **(Oral/Research)** Ask at home: "Where does the water we drink in this house come from?" (e.g., Borehole, Well, Tap from government).

WEEK 5: MIDTERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

Attempt questions covering topics from Week 1 to Week 4 correctly.

CONTENT

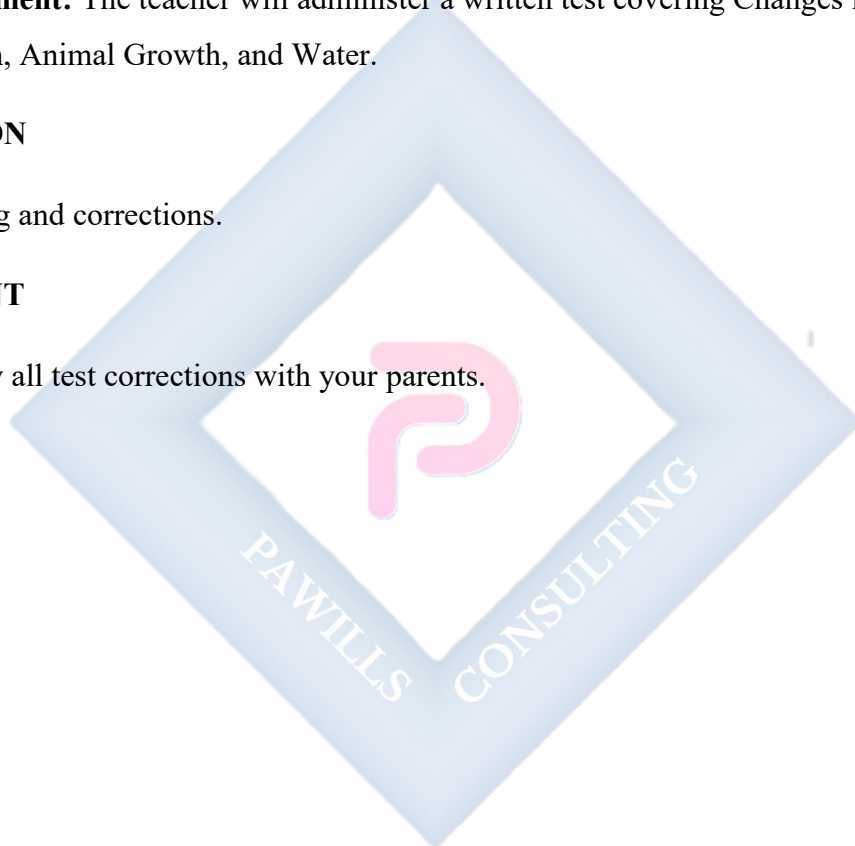
- **Assessment:** The teacher will administer a written test covering Changes in Nature, Plant Growth, Animal Growth, and Water.

EVALUATION

- Grading and corrections.

ASSIGNMENT

- Review all test corrections with your parents.



WEEK 6: MIDTERM BREAK



WEEK 7: HUMAN BODY: THE MOUTH

OBJECTIVES

By the end of the lesson, pupils should be able to:

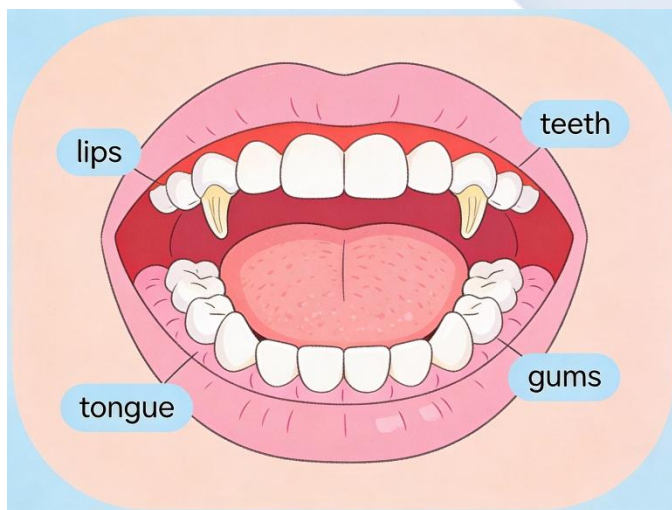
1. Identify the parts of the mouth (lips, teeth, tongue, gums).
2. State the function of the tongue.
3. Explain the importance of the mouth for eating and speaking.
4. Mention two ways to care for the mouth.

CONTENT

Parts of the Mouth

The mouth is the opening on our face we use for eating and speaking. Let's look inside:

- **Lips:** The soft, movable outer part. They open and close. They help us hold food in and form words like "p" and "b".
- **Teeth:** The hard, white structures inside. They are for biting, cutting, and chewing food into small pieces.
- **Tongue:** The soft, flexible, pink muscle that can move in many directions. It is very strong!
- **Gums:** The pink, soft flesh that surrounds and holds the teeth firmly in place.



Functions of the Mouth

Our mouth is a very busy part of our body!

1. **Eating (Ingestion):** Food enters our body through the mouth.
2. **Chewing (Mastication):** Teeth grind solid food (like meat or gari) into a soft paste that is easy to swallow and digest.
3. **Tasting:** The top of our tongue has tiny bumps called taste buds. They help us know if food is Sweet (like sugar), Sour (like lemon), Salty (like salt), or Bitter (like bitter leaf).
4. **Speaking (Talking):** The lips, tongue, and teeth work together with our voice box to make different sounds and form words. Try saying "la la la" and see your tongue move!

Oral Hygiene (Care of the Mouth)

If we don't take care of our mouth, it can smell bad (halitosis) and our teeth can get holes (cavities).

- Brush your teeth twice a day – when you wake up and before you go to bed.
- Rinse your mouth with water after eating, especially after sweets.
- Do not use your teeth to break hard things like bones, bottle caps, or ice.
- Visit a dentist if you have a toothache.

EVALUATION

1. Name three parts you can see when you open your mouth wide.
2. Which part of the mouth tells you if your food is sweet or salty?
3. What is the main job of the teeth?
4. Can we speak clearly if we have food in our mouth? Why?
5. How many times should we brush our teeth every day?

ASSIGNMENT

1. **(Drawing)** Draw a happy face with a big smile, showing lips and teeth.
2. **(Written)** Write down two different tastes you experienced today (e.g., Sweet - tea, Salty - soup).

3. **(Oral/Reflection)** What do you think will happen if you never brush your teeth?



WEEK 8: HUMAN BODY: THE TEETH

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Differentiate between milk teeth and permanent teeth.
2. List the types of teeth (Incisors, Canines, Premolars, Molars).
3. State the function of each type of tooth.
4. Mention two common dental problems.

CONTENT

Sets of Teeth

In our lifetime, we grow two sets of teeth:

1. **Milk Teeth (Baby Teeth):** The first set of 20 teeth. They start growing when a baby is about 6 months old. They are smaller, whiter, and not as strong. They begin to fall out from about age 6 to make way for the second set.
2. **Permanent Teeth:** The second and final set. There are 32 of them. They are bigger, stronger, and slightly less white. They must last us for our whole life, so we must take very good care of them.



Milk Teeth

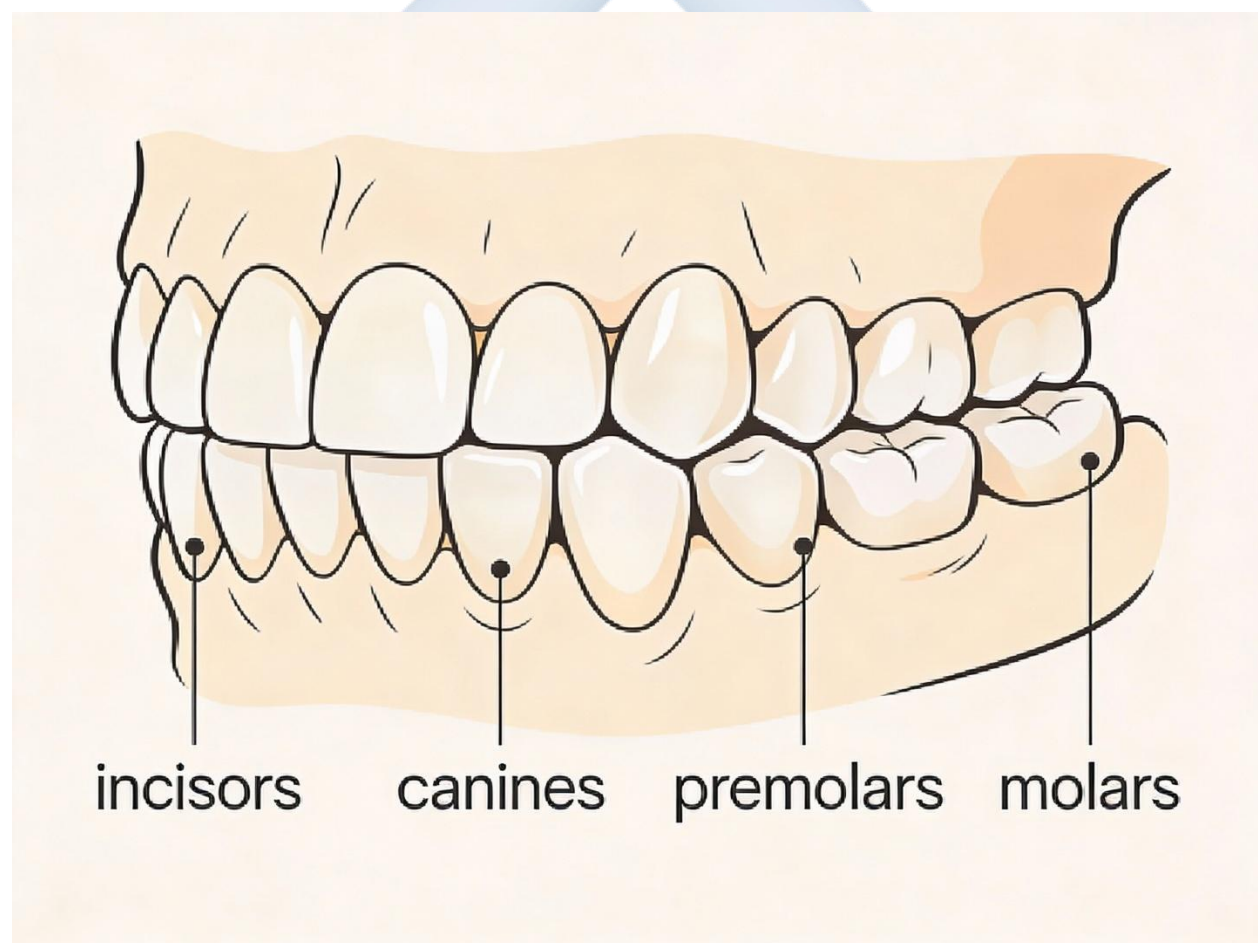


**Permanent
Teeth**

Types of Teeth and Their Functions

Our teeth are not all the same shape. They have different jobs, just like tools in a toolbox.

Type of Tooth	Where it is	Shape	Job (Function)	Example of Food
Incisors	Front of mouth	Flat, sharp edge	Cutting and biting off pieces of food	Apple, bread
Canines	Beside incisors	Pointed, sharp	Tearing and ripping food	Meat, tough meat
Premolars & Molars	Back of mouth	Broad, flat with bumps	Grinding, crushing, and chewing food into paste	Yam, rice, meat



Dental Care and Problems

- **Common Problems:**

- **Tooth Decay (Cavities):** Holes in the teeth caused by germs that eat sugar and produce acid. It starts as a black spot and can cause pain (toothache).
- **Toothache:** Severe pain in or around a tooth.

- **How to Care for Teeth:**

- Brush properly twice daily with fluoride toothpaste.
- Eat healthy foods like fruits, vegetables, and milk which make teeth strong.
- Limit sweets, chocolates, and sugary drinks.
- See a dentist for check-ups.

EVALUATION

1. How many sets of teeth do humans have in their life? What are they called?
2. Which teeth are shaped like chisels and are used for cutting?
3. If you are eating a piece of tough meat, which teeth do you mainly use to tear it?
4. What causes black holes (cavities) to form in our teeth?
5. Name one food that is good for making your teeth strong.

ASSIGNMENT

1. **(Practical/Observation)** Look in a mirror, open your mouth wide, and count your front teeth (top and bottom). Write down the number.
2. **(Drawing)** Draw the shape of a broad, flat molar tooth.
3. **(Written)** List two bad habits that can spoil your teeth (e.g., eating too much sweet, using teeth to open bottles).

WEEK 9: NON-LIVING THINGS: ROCKS

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define rocks as hard, non-living things.
2. List three characteristics of rocks (hard, solid, different colors).
3. Mention three uses of rocks in daily life.
4. Identify rocks in the school environment.

CONTENT

Characteristics of Rocks

Rocks are hard, solid pieces of the earth's crust. They are **non-living things**.

- **They do not have life:** They cannot grow, breathe, eat, reproduce, or die like plants or animals.
- **They are hard and solid:** You cannot squeeze or bend a rock. They are strong.
- **They come in different sizes:** From tiny **pebbles** (like gravel), to medium **stones**, to huge **boulders**.
- **They have different colours and textures:** Some are grey (like granite), some black, white, or even red. Some are smooth (river stones), some are rough and jagged.



Common Types of Rocks Around Us

- **Granite:** Very hard, grey, speckled rock. Used for building.
- **Limestone:** Used to make cement.
- **Sandstone:** Feels sandy. Sometimes used for grinding.
- **Laterite:** Reddish-brown soil that hardens like rock. Used in some local buildings.

Uses of Rocks Rocks are very useful in our daily lives.

1. **Building and Construction:** Crushed rock (gravel) is mixed with sand and cement to make concrete for houses, bridges, and roads. Slabs of granite are used for kitchen counters.
2. **Decoration:** Beautiful, shiny rocks and stones are used in jewelry (beads), to decorate gardens, and in fish tanks.
3. **Tools:** In the olden days and villages, hard rocks were shaped into tools for grinding grains, pepper, and melon.
4. **Other Uses:** Chalk used on blackboards is made from a soft rock. Some rocks contain metals like iron that we use to make cars.

EVALUATION

1. Is a rock a living thing or a non-living thing? Give one reason for your answer.
2. Mention two ways you can describe a rock (e.g., by its colour or how it feels).
3. What do builders mix with cement and sand to make strong concrete?
4. Can a small pebble grow into a big boulder by itself?
5. Look around your school compound. Where can you find rocks or stones?

ASSIGNMENT

1. **(Practical/Collection)** Find and collect three small stones from around your home. Bring them to class. Describe one (e.g., "smooth and grey").
2. **(Written)** Look around your house. List two things that are made from or contain rock/stone (e.g., the foundation of the house, a grinding stone, a decorative stone in a flower pot).

3. **(Drawing)** Draw a picture of a big, strong rock or boulder.



WEEK 10: PROPERTIES OF LIVING AND NON-LIVING THINGS

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Distinguish between living and non-living things.
2. List four characteristics of living things (MR NIGER D).
3. Explain movement and nutrition in living things.
4. Give examples of living and non-living things.

CONTENT

Differences Between Living and Non-Living Things

Everything around us can be put into two big groups: **Living** and **Non-Living**.

- **Living Things:** Have life in them. They can do certain activities by themselves.
Examples: Human being, dog, lizard, mango tree, grass.
- **Non-Living Things:** Do not have life. They cannot do things on their own; they need to be moved or used by living things. *Examples:* Stone, book, bicycle, water, shirt.

Characteristics of Living Things (MR NIGER D)

How do we know something is alive? We use the code **MR NIGER D** to remember the eight features of all living things.

1. **M - Movement:** Living things can move. Animals walk, run, fly. Plants move slowly (roots grow down, leaves turn towards the sun).
2. **R - Respiration:** Living things breathe. They take in air (oxygen) and give out air (carbon dioxide).
3. **N - Nutrition:** Living things need food. Food gives them energy to grow and work.
4. **I - Irritability (Sensitivity):** Living things respond to changes. If you touch a hot pot (change), you quickly move your hand (response). A plant grows towards light.
5. **G - Growth:** Living things grow. A puppy becomes a bigger dog. A seedling becomes a tall tree.

6. **E - Excretion:** Living things produce waste. We sweat and urinate. Plants release oxygen as waste during the day.
7. **R - Reproduction:** Living things can make young ones like themselves. A cat gives birth to kittens. A plant produces seeds.
8. **D - Death:** All living things eventually die when their life ends.

MR NIGER D Comparison Chart		
Living (Cat)	Non-Living (Rock)	
Movement	☒	☐
Respiration	☒	☐
Nutrition	☒	☐
Growth	☒	☐
Excretion	☒	☐
Reproduction	☒	☐
Sensitivity	☒	☐



Non-Living Things They do **not** have the features of MR NIGER D. A car can move (because of fuel and a driver) but it cannot grow, breathe, or have babies. Therefore, it is non-living.

EVALUATION

1. Is a table a living thing? Give two reasons why it is not.
2. What does the 'N' in MR NIGER D stand for? Explain it simply.
3. A car moves. Is it a living thing? Why or why not?
4. Give two examples of living things you can find in a farm.
5. Give two examples of non-living things you can find in a classroom.

ASSIGNMENT

1. **(Written)** Write down the full meaning of MR NIGER D.

2. **(Drawing)** Draw one living thing (e.g., a bird) and one non-living thing (e.g., a ball) in your notebook.
3. **(Reasoning)** Does a plant move? Explain your answer with an example (think about its parts).



WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts taught from Week 1 to Week 10.
2. Answer questions on Changes in Nature, Human Body, and Living Things.

CONTENT

Comprehensive Review

- **Nature:** Day/Night, Seasons (Dry/Rainy), Weather (Sunny, Rainy, Cloudy, Windy).
- **Water:** Sources, uses, and conservation.
- **Biology:** Plant growth (Germination, Stages), Animal life cycles (Chicken, Butterfly), Human growth stages.
- **Human Body:** Parts and care of the Mouth and Teeth.
- **Science:** Rocks (Non-living things), Characteristics of Living Things (MR NIGER D).

EVALUATION

- Oral quiz and written revision exercises covering the term's work.

ASSIGNMENT

- Study your notes from Week 1 to Week 10 for the end-of-term examination.

PRIMARY 4 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	Light Energy	Sources of light - natural and artificial; Properties of light; Light and shadows; Reflection of light; Importance of light
2	Basic Electricity	Sources of electrical energy; Electrical appliances in the home; Conductors and insulators; Safety with electricity; Simple circuits
3	Sound Energy	Sources of sound; How sound travels; Loud and soft sounds; Musical instruments; Noise pollution
4	Heat Energy	Sources of heat; Effects of heat on materials; Heat and temperature; Uses of heat; Safety with hot objects
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Air and Breathing	Composition of air; Importance of air; The breathing system; Keeping lungs healthy; Air pollution
8	Soil as Part of Our Environment	Types of soil; Properties of different soils; Uses of soil; Soil conservation; Organisms in the soil
9	The Solar System	The sun, moon and stars; Planets in the solar system; Day and night; The earth's movement
10	Weather and Climate	Elements of weather; Weather instruments; Seasons in Nigeria; Dressing for different weather conditions
11	REVISION	Review of all topics covered; Practice exercises; Question and answer sessions
12	EXAMINATION	End of term assessment
13	CLOSING	Report card distribution; Vacation assignment

PRIMARY 4 BASIC SCIENCE AND TECHNOLOGY LESSON NOTES

SECOND TERM

WEEK 1: LIGHT ENERGY

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define light energy.
2. List three natural and three artificial sources of light.
3. Explain how shadows are formed.
4. State two uses of light in our daily lives.

CONTENT

What is Light Energy?

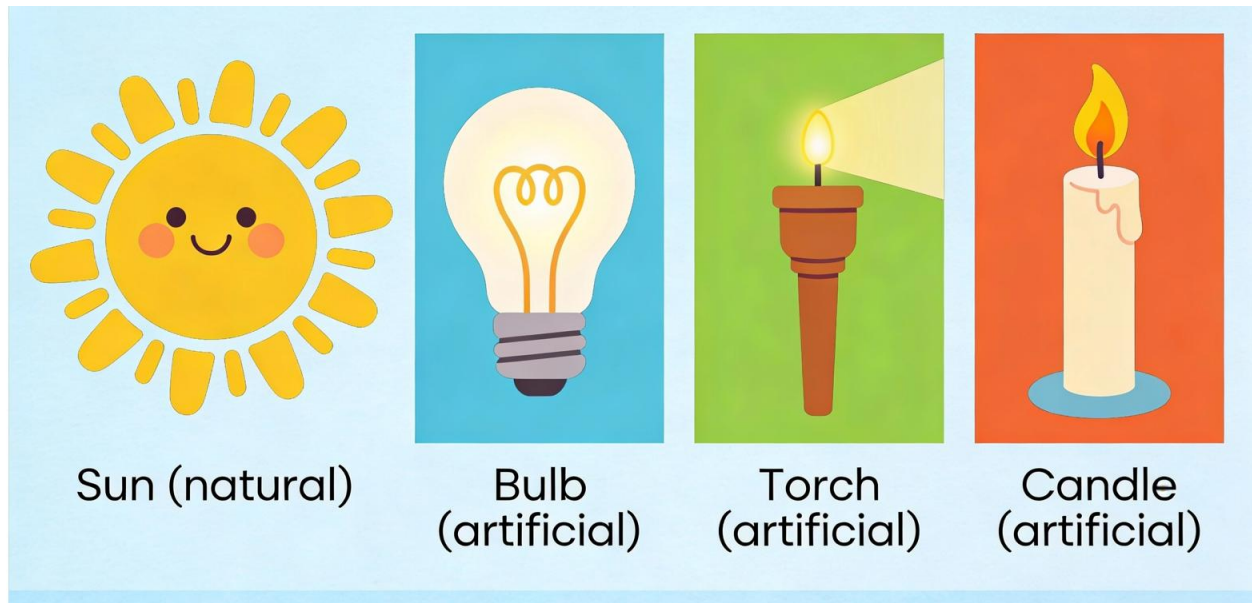
Light is a form of energy that our eyes can detect. It makes things visible. Without light, we would be in complete darkness and unable to see anything.

Sources of Light

Where does light come from?

- **Natural Sources:** Given to us by nature. They are not made by man.
 - **The Sun:** The biggest and most important source. It gives us daylight.
 - **The Moon:** It does not make its own light. It reflects the Sun's light like a mirror at night.
 - **Stars:** They are like very far-away suns, twinkling at night.
 - **Fireflies:** Small insects that produce their own tiny light.
- **Artificial Sources:** Made by humans because we need light when natural light is gone (at night or inside dark rooms).
 - **Electric Bulbs/Tubes:** In our homes and schools.
 - **Torchlight (Flashlight):** Uses batteries.

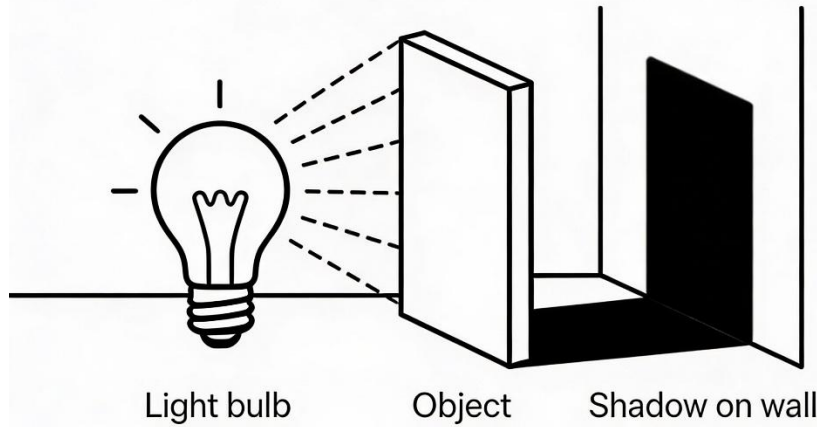
- **Candle, Kerosene Lamp, Firewood:** Used where there is no electricity.



Properties of Light and Shadows

- **Light Travels in Straight Lines:** Light rays move in straight paths. If you shine a torch in a dark room, you see a straight beam, not a 弯曲的 one. This is why light cannot go around corners.
- **Formation of Shadows:** A shadow is a dark area that forms when an object blocks light. Because light travels straight, it cannot bend to go behind the object, leaving a dark shape on the wall or ground.
 - Bigger objects make bigger shadows. Moving the light source changes the shadow's size and position.
- **Reflection of Light:** When light hits a smooth, shiny surface (like a mirror, calm water, or shiny metal), it bounces back. This is called reflection. That's how we see ourselves in a mirror.

Shadow Formation Diagram



Importance of Light

1. **For Sight (Vision):** The most important use. It allows us to see our surroundings, read, write, and recognize people.
2. **For Plants:** Plants need sunlight to make their own food through a process called photosynthesis. No light, no food, plant dies.
3. **For Safety:** Street lights and car headlights help us see at night to avoid accidents and criminals.
4. **For Other Uses:** Drying clothes, taking photographs, generating electricity (solar panels).

EVALUATION

1. What is the name of the main natural light source we have during the day?
2. Name two things in your house that give artificial light.
3. How can you create a shadow of your hand on the wall?
4. Why can't light help you see around a corner in a dark hallway?
5. Why would a plant kept in a dark cupboard not grow well?

ASSIGNMENT

1. **(Drawing)** Draw a picture of a candle with light rays (lines) coming from its flame.
2. **(Practical/Observation)** Go outside in the sunlight around noon. Look at your shadow. Is it long or short? What colour is it? Write down your observation.
3. **(Written)** Make a list of three different sources of light you used from evening until you went to bed last night.



WEEK 2: BASIC ELECTRICITY

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define electricity.
2. List three sources of electricity.
3. Identify electrical appliances used at home.
4. Differentiate between conductors and insulators.

CONTENT

What is Electricity?

Electricity is a very useful form of energy that can flow through wires. It makes our lives easier by powering many machines and giving us light.

Sources of Electricity

How do we get electricity?

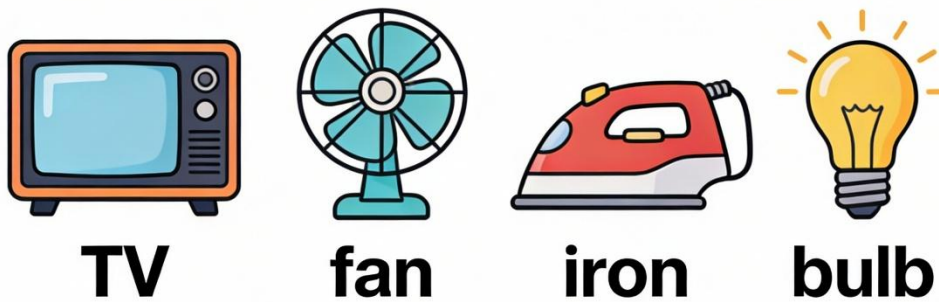
- **Batteries (Dry Cells):** Small, portable sources. Used in remote controls, toys, torches, and wall clocks.
- **Generators:** Machines that use petrol or diesel fuel to produce electricity. Common in Nigeria when there is no mains power.
- **Mains Supply (PHCN/IKEJA Electric, etc.):** Electricity generated at big power stations and sent to our homes through overhead cables and underground wires.
- **Solar Panels:** Special plates that convert sunlight directly into electricity. Very useful in sunny places.

Electrical Appliances

These are devices that need electricity to work. We plug them into sockets or use batteries.

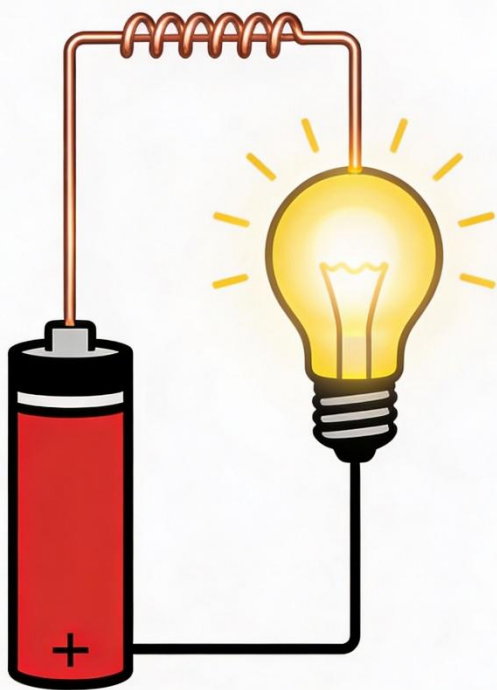
- **At Home:** Television, Refrigerator, Electric Iron, Blender, Fan, Air Conditioner, Electric Kettle.

- **At School:** Computers, Projectors, Electric Bells.
- **For Light:** Bulbs, Fluorescent tubes, Lamps.

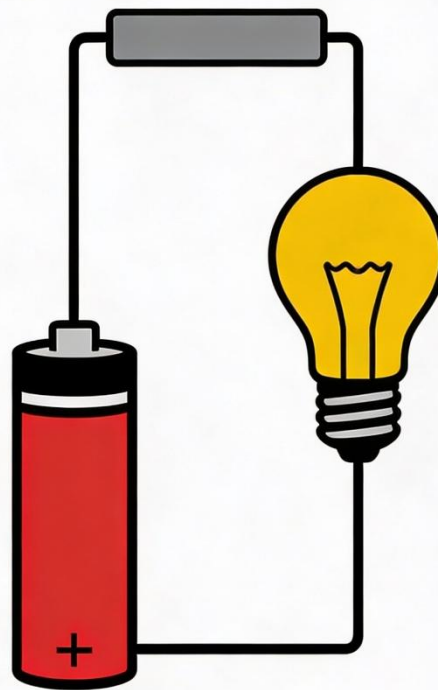


Conductors and Insulators

- **Conductors:** Materials that allow electric current to pass through them easily. **Metals** are good conductors.
 - *Examples:* Copper wire (inside cords), Iron, Aluminium, Water, and the human body. **This is why electricity is DANGEROUS.**
- **Insulators:** Materials that **do not** allow electric current to pass through easily. They stop electricity.
 - *Examples:* Plastic, Rubber, Wood, Glass, Dry cloth. This is why electric wires are covered with plastic or rubber – to protect us from shock.



Conductor Circuit



Insulator Circuit

Electrical

Safety Rules

- **NEVER** touch electrical sockets, plugs, or switches with wet hands.
- **NEVER** put fingers or objects (like nails, keys) into plug holes.
- Tell an adult if you see a broken wire or a sparking socket.
- Do not climb electric poles or fly kites near power lines.

EVALUATION

1. What do we call the small, square objects that make a torchlight work?
2. Name two appliances in your home that you must plug into a wall socket to use.
3. What is a conductor? Give one example.
4. Why is the outside of an electric wire made of plastic?
5. If you need to hold a metal part of a plugged-in appliance, should your hand be wet or dry? Why?

ASSIGNMENT

1. **(Drawing)** Draw the shape of a common AA battery and label the positive (+) and negative (-) ends.
2. **(Written)** Go around your sitting room. List five things that use electricity (either by plug or battery).
3. **(Reasoning)** Explain to a younger sibling: Why is it dangerous to touch a TV plug with wet hands?



WEEK 3: SOUND ENERGY

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define sound energy.
2. Mention three sources of sound.
3. Explain simply how sound travels.
4. Differentiate between loud and soft sounds.
5. Define noise pollution.

CONTENT

What is Sound?

Sound is a form of energy that we hear with our **ears**. Sound is produced when an object vibrates (moves back and forth very quickly).

Sources of Sound

Anything that vibrates can make sound.

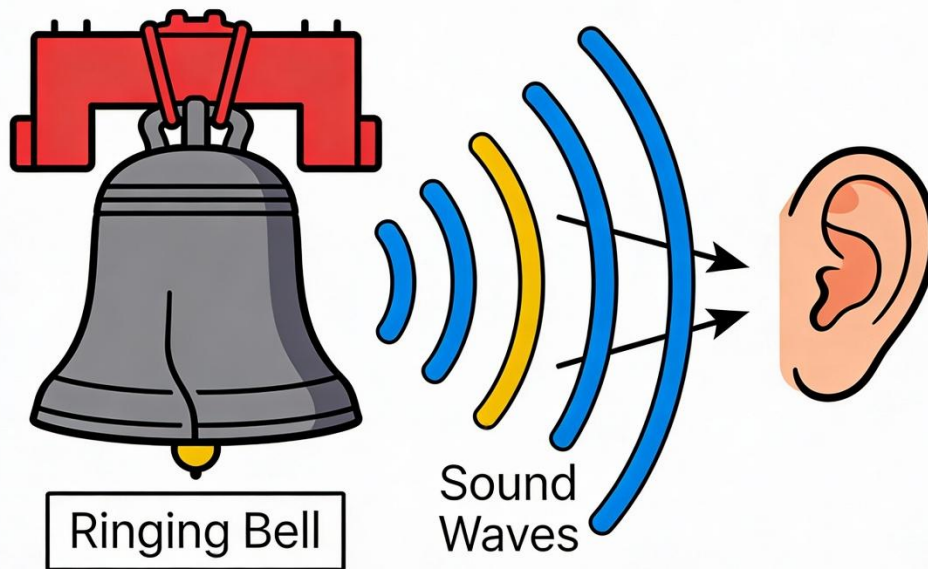
- **Living Things:** People talking, singing, clapping. Dogs barking, birds chirping, cows mooing.
- **Musical Instruments:** Drum (vibrating skin), Guitar (vibrating string), Flute (vibrating air column).
- **Machines and Vehicles:** Car horn, generator sound, aeroplane engine, ringing phone, school bell.
- **Nature:** Thunder, wind blowing through trees, flowing water.

How Sound Travels

Sound needs a **medium** (material) to travel through. It cannot travel through empty space (vacuum).

- It travels fastest through **solids** (put your ear on a desk and tap it), then **liquids**, then **gases** (like air).
- When a drum is hit, the skin vibrates and pushes the air particles next to it. These particles bump into the next ones, creating a sound wave that travels to your ear.

Sound Waves Diagram



Loud and Soft Sounds

- **Loud Sounds:** Have a lot of energy. They are caused by strong vibrations. *Examples:* Thunder, aeroplane taking off, loud music from speakers, shouting.
- **Soft Sounds:** Have less energy. They are caused by gentle vibrations. *Examples:* Whispering, rustling of leaves, ticking of a clock, a cat purring.

Noise Pollution

- **Noise** is any unpleasant, disturbing, or unwanted sound.
- **Noise Pollution** is when there is too much loud, unpleasant sound in the environment for a long time.
- **Effects:** It can cause headache, disturb sleep, make it hard to concentrate, and over time, damage hearing.

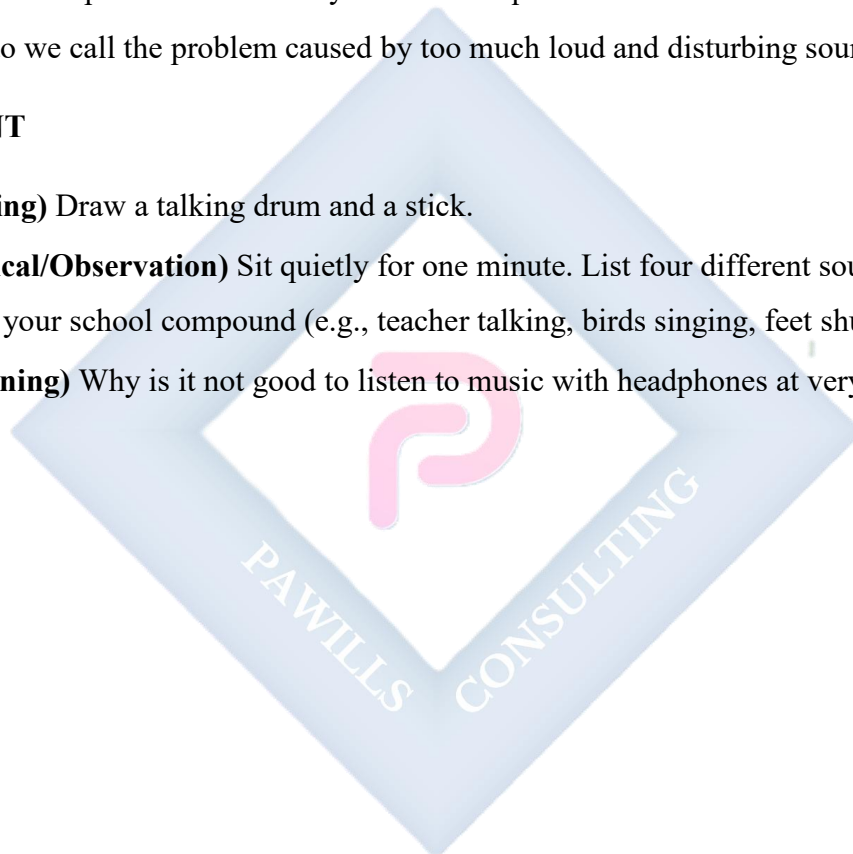
- **Sources:** Constant loud generators, loud street music (okada speakers), traffic honking, noise from construction sites.

EVALUATION

1. Which part of our body do we use to hear sound?
2. How is sound created? (What must happen to an object?)
3. Give an example of a loud sound you hear on your street.
4. Give an example of a soft sound you hear in a quiet classroom.
5. What do we call the problem caused by too much loud and disturbing sound in our area?

ASSIGNMENT

1. **(Drawing)** Draw a talking drum and a stick.
2. **(Practical/Observation)** Sit quietly for one minute. List four different sounds you can hear in your school compound (e.g., teacher talking, birds singing, feet shuffling).
3. **(Reasoning)** Why is it not good to listen to music with headphones at very high volume?



WEEK 4: HEAT ENERGY

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define heat energy.
2. List three sources of heat.
3. State three uses of heat in our daily life.
4. Mention the difference between heat and temperature.

CONTENT

What is Heat Energy?

Heat is a form of energy that makes things hot. We can feel heat with our skin. It flows from a hotter object to a cooler one.

Sources of Heat

- **The Sun:** The biggest natural source. It heats up the Earth.
- **Fire:** Produced by burning materials like wood, charcoal, kerosene, gas, or petrol.
- **Electricity:** Many electrical appliances produce heat as their main job or as a side effect.
Examples: Electric cooker, Iron, Heater, Toaster, Bulb (gets hot).
- **Friction:** When two surfaces rub together, they get warm. Rub your palms together very fast for 30 seconds. They feel warm!

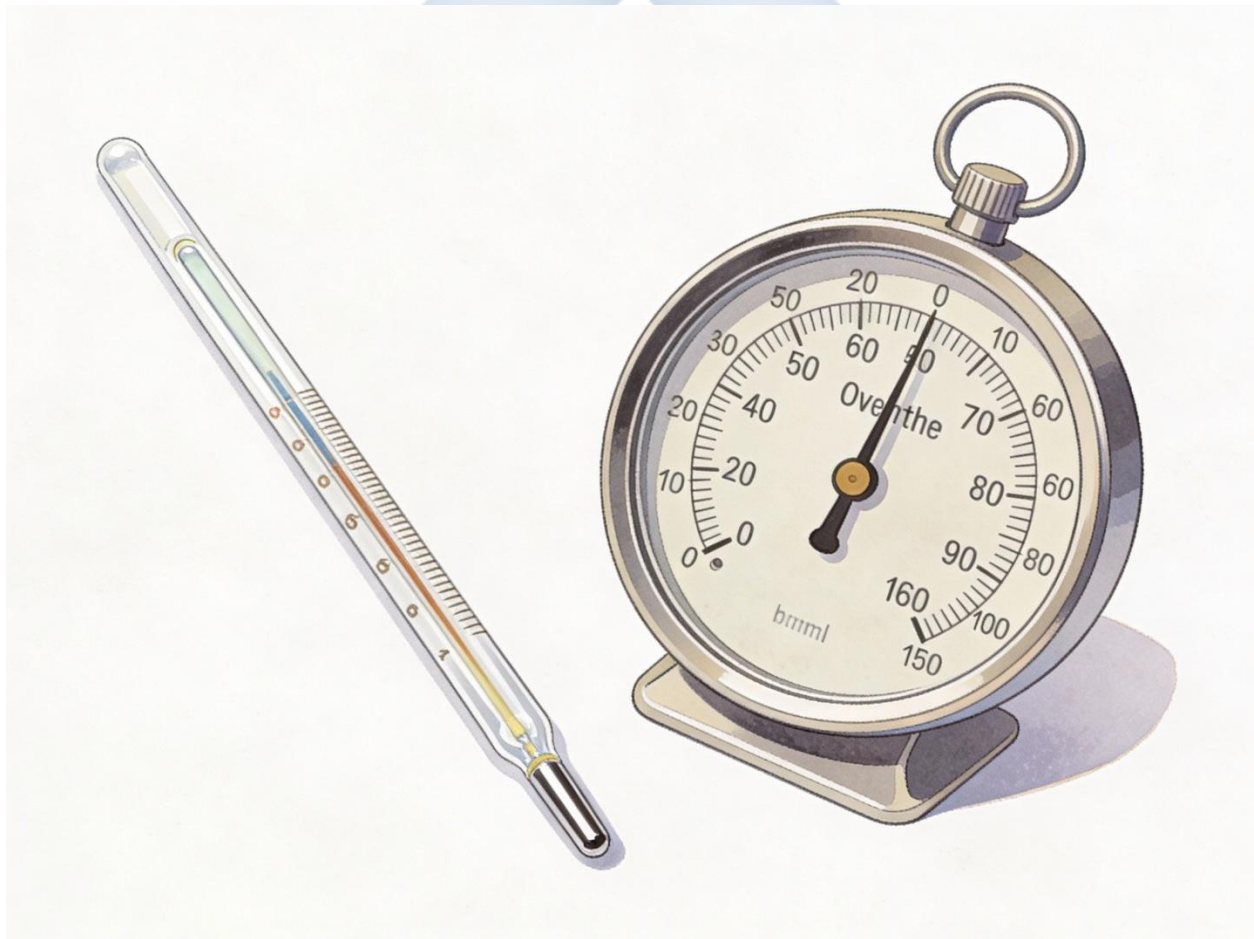
Uses of Heat Heat is very useful in our daily activities:

1. **Cooking:** Heat cooks our food, making it soft, tasty, and safe to eat (kills germs). We boil, fry, roast, and bake.
2. **Drying:** The sun's heat dries wet clothes on the line. It also dries crops like maize, pepper, and fish after harvest.
3. **Ironing:** Heat from an iron removes wrinkles from clothes, making them neat.
4. **Keeping Warm:** In cold weather or during Harmattan, we use heaters, wear thick clothes, or sit near a fire to feel warm.

5. **Other Uses:** Melting ice, shaping metals (welding), generating electricity in power stations.

Heat vs. Temperature

- **Heat:** Is the **energy** itself. It is the total amount of hotness.
- **Temperature:** Is a **measure** of how hot or cold something is. It tells us the *degree* of hotness.
 - We measure temperature with an instrument called a **thermometer**. It can be in degrees Celsius ($^{\circ}\text{C}$) or Fahrenheit ($^{\circ}\text{F}$).
 - *Example:* A cup of tea and a bucket of water might have the same **temperature** (e.g., 80°C), but the bucket has more **heat energy** because there is more hot water.



Safety with Heat

- Do not play with matches, lighters, or candles.
- Use a pot-holder or dry cloth to handle hot pots and pans.
- Be careful around cooking stoves and heaters.
- Do not touch a hot electric bulb.

EVALUATION

1. What is the biggest natural source of heat for the Earth?
2. Mention two ways your mother uses heat in the kitchen.
3. What happens when you rub your hands together quickly? What is this source of heat called?
4. What is the name of the instrument used to check if you have fever?
5. Why should you not touch a hot pressing iron with your bare hands?

ASSIGNMENT

1. **(Drawing)** Draw a pot sitting on a burning firewood stove (local or gas cooker).
2. **(Written)** Ask your mother or father: List three different things they used heat for while preparing yesterday's dinner.
3. **(Practical/Observation)** Rub your palms together firmly and fast for one full minute (count to 60). Stop and immediately feel your palms. Write down what you felt.

WEEK 5: MIDTERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

Attempt questions covering topics from Week 1 to Week 4 correctly.

CONTENT

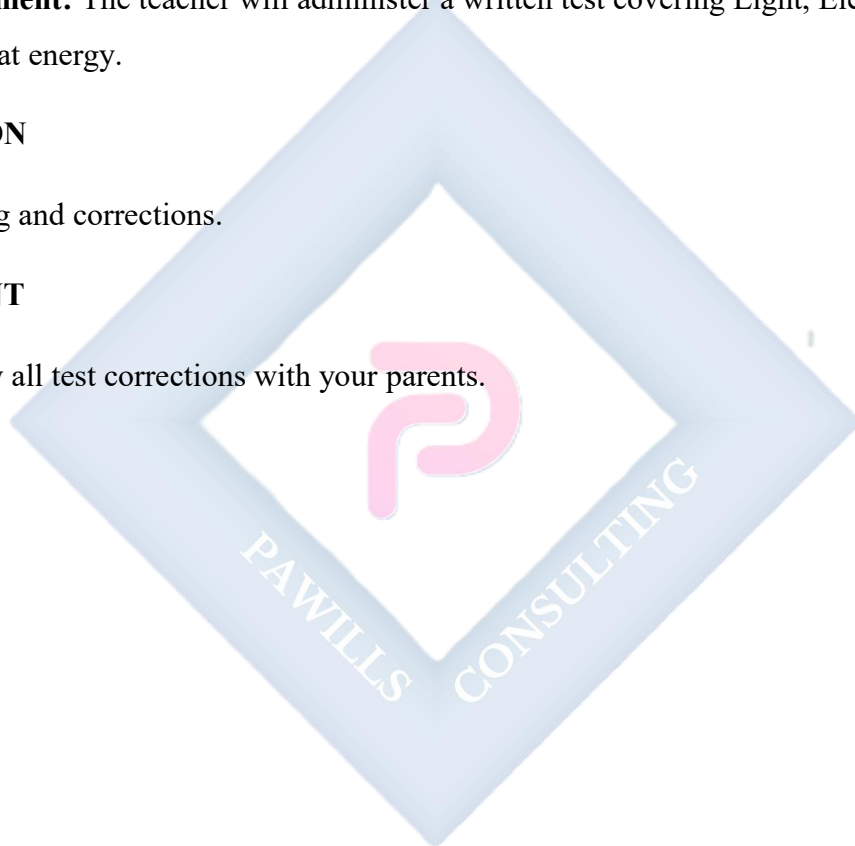
- **Assessment:** The teacher will administer a written test covering Light, Electricity, Sound, and Heat energy.

EVALUATION

- Grading and corrections.

ASSIGNMENT

- Review all test corrections with your parents.



WEEK 6: MIDTERM BREAK



WEEK 7: AIR AND BREATHING

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define air.
2. List the gases found in air (Oxygen, Carbon dioxide, Nitrogen).
3. Explain the importance of air to living things.
4. Identify the main organ for breathing (Lungs).

CONTENT

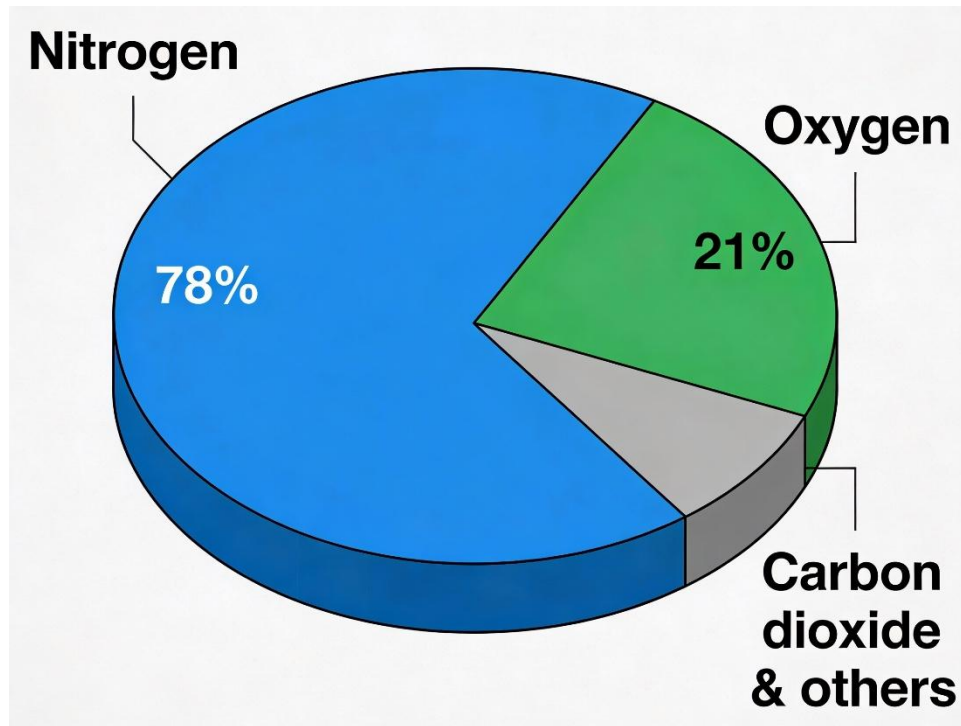
What is Air?

Air is the mixture of gases that surrounds the Earth. We cannot see it, but we can feel it when it moves (wind). It is everywhere – in your classroom, in water bubbles, and in soil spaces.

Composition of Air (What Air is Made Of)

Air is not just one thing; it is a mixture. The main gases are:

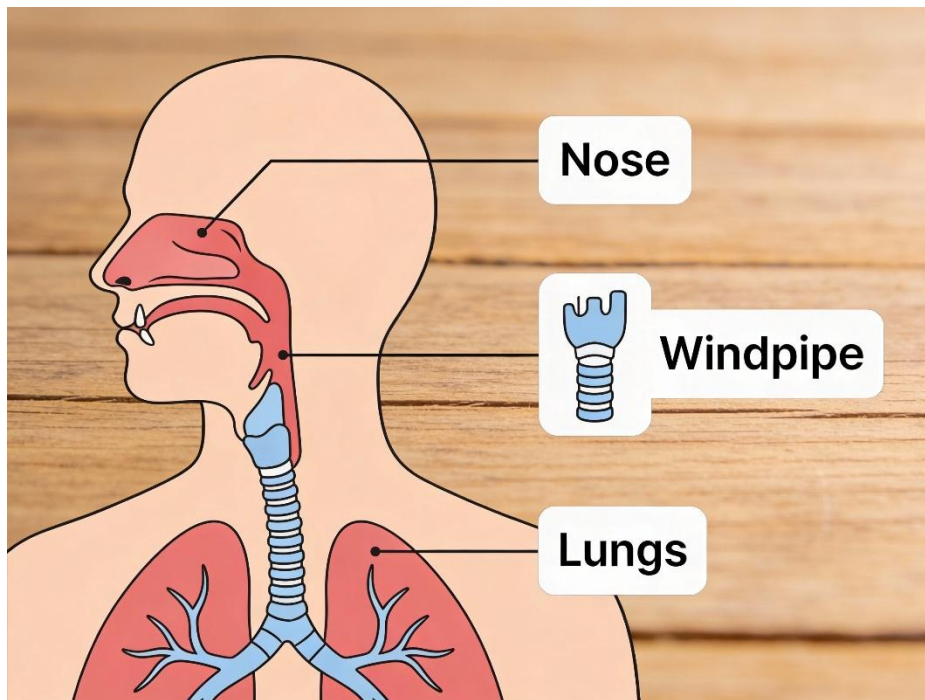
- **Nitrogen (N₂):** Makes up about **78%** (more than three-quarters) of the air. Most plants need it to grow.
- **Oxygen (O₂):** Makes up about **21%** (just over one-fifth). This is the gas that humans and animals **breathe in** to stay alive. It is used to burn the food in our bodies to give us energy.
- **Carbon Dioxide (CO₂):** Makes up a very small amount (about 0.04%). Humans and animals **breathe this out**. Plants need it to make their food (photosynthesis).
- **Other gases:** There are tiny amounts of other gases and water vapour.



The Breathing System

Breathing is the process of taking in air (inhaling) and giving out air (exhaling).

1. We breathe in air through our **Nose** (or mouth). The nose has hairs that clean the air.
2. The air goes down a tube called the **Windpipe**.
3. The windpipe divides into two tubes that lead to the **Lungs**.
4. The **Lungs** are like two spongy bags inside our chest. They take the **oxygen** from the air and put it into our blood. The blood carries oxygen to all parts of the body. At the same time, the lungs remove waste **carbon dioxide** from the blood and we breathe it out.



Keeping Lungs Healthy and Air Pollution

- **For Healthy Lungs:** Exercise (run, play), breathe fresh air, and avoid smoking.
- **Air Pollution:** When harmful things get into the air and make it dirty and unsafe to breathe.
 - **Causes:** Smoke from generators, cars, factories, burning refuse, dust, and cigarette smoke.
 - **Effects:** Causes coughing, asthma, and can make people sick.

EVALUATION

1. Can we see the air around us? How do we know it is there?
2. Which gas in the air is most important for us to breathe in to stay alive?
3. Which gas do we produce as waste and breathe out?
4. What is the name of the main organ in our chest that helps us to breathe?
5. Mention one human activity that makes the air dirty (pollutes it).

ASSIGNMENT

1. **(Drawing)** Draw the outline of the human body from the neck to the stomach. Draw two bag-like shapes in the chest and label them "LUNGS".
2. **(Practical/Observation)** Place your hand on your chest. Take a deep breath in. What happens to your chest? Does it go up or down? Write your answer.
3. **(Written)** Name two places in your community where the air often smells bad or looks smoky/dirty.



WEEK 8: SOIL AS PART OF OUR ENVIRONMENT

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define soil.
2. Identify three types of soil.
3. State the properties of Sandy, Clayey, and Loamy soil.
4. Mention the best soil for farming.

CONTENT

What is Soil?

Soil is the loose, upper layer of the earth's surface where plants grow. It is a mixture of:

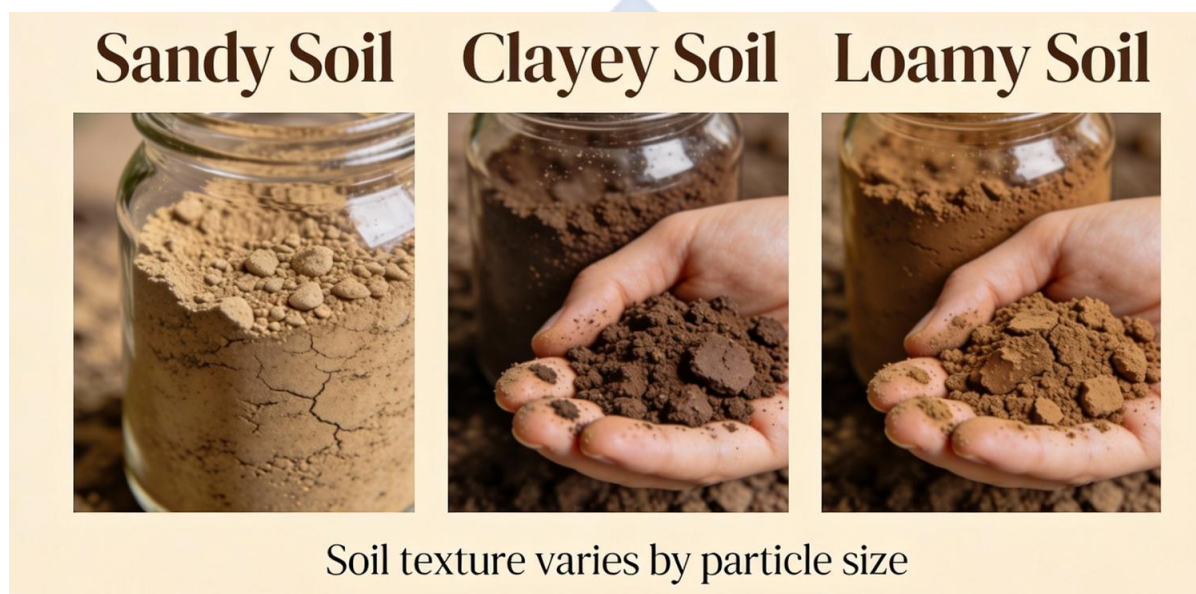
- **Broken pieces of rock** (from big rocks breaking down over many years).
- **Decayed plants and animals** (called **humus** – this makes soil fertile, dark and rich).
- **Water and Air** (in the spaces between soil particles).
- **Living organisms** like earthworms, ants, and bacteria.

Types of Soil and Their Properties

There are three main types:

Type of Soil	How it Feels/Looks	Water Holding Ability	Best For...
Sandy Soil	Gritty, rough, loose. Particles are large.	Poor. Water drains through very quickly. Dries fast.	Not good for most crops. Used in construction. Found at beaches.
Clayey Soil	Smooth, sticky when wet. Hard and cracked when dry. Particles are very fine.	Very Good (Too Good). Holds too much water, becomes waterlogged.	Making pots, bricks, and pottery. Not good for plants that need good drainage.

Type of Soil	How it Feels/Looks	Water Holding Ability	Best For...
Loamy Soil	Dark, crumbly, soft. A mixture of sand, clay, and lots of humus.	Excellent. Holds just the right amount of water and air.	THE BEST FOR FARMING. Most crops like maize, vegetables, yam grow very well in it.



Importance of Soil

- **For Plants:** It provides support, water, air, and nutrients (food) for plants to grow.
- **For Animals:** Many small animals (earthworms, insects, rodents) live in the soil.
- **For Humans:** We grow our food in soil. We also use clay soil to build houses (mud bricks) and make pots.

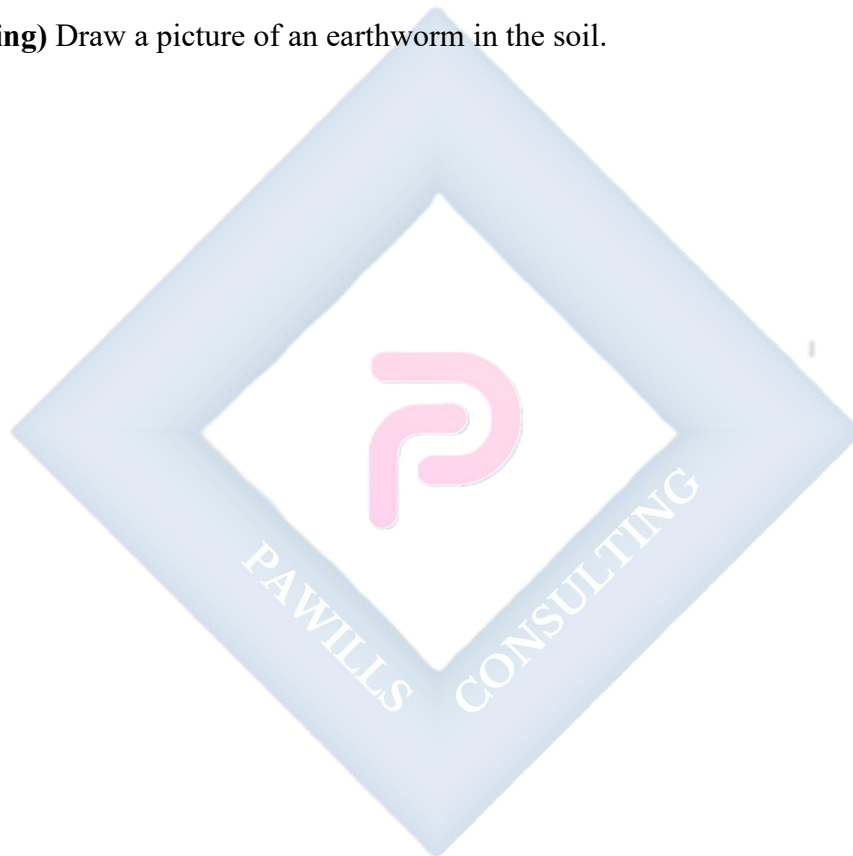
EVALUATION

1. Name the three main types of soil.
2. Which type of soil feels sticky when it is wet?
3. If a farmer wants to plant vegetables, which soil should he look for? Why?
4. Name one small animal that lives in the soil and helps to loosen it.

5. Why is sandy soil not good for planting flowers in a pot? (Hint: Think about water).

ASSIGNMENT

1. **(Practical/Collection)** With the help of an adult, collect small samples (a handful each) of sand, clay, and dark garden soil. Bring them to class in separate small bags or containers.
2. **(Experiment)** Take a small amount of clay soil and sandy soil. Add a few drops of water to each. Which one becomes sticky? Write down your observation.
3. **(Drawing)** Draw a picture of an earthworm in the soil.



WEEK 9: THE SOLAR SYSTEM

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify the components of the solar system (Sun, Planets, Moon).
2. State that there are eight planets.
3. Explain simply how day and night occur.
4. Mention which planet we live on.

CONTENT

The Solar System

The Solar System is like a giant family in space, with the Sun as the father at the center. All the other members move around the Sun.

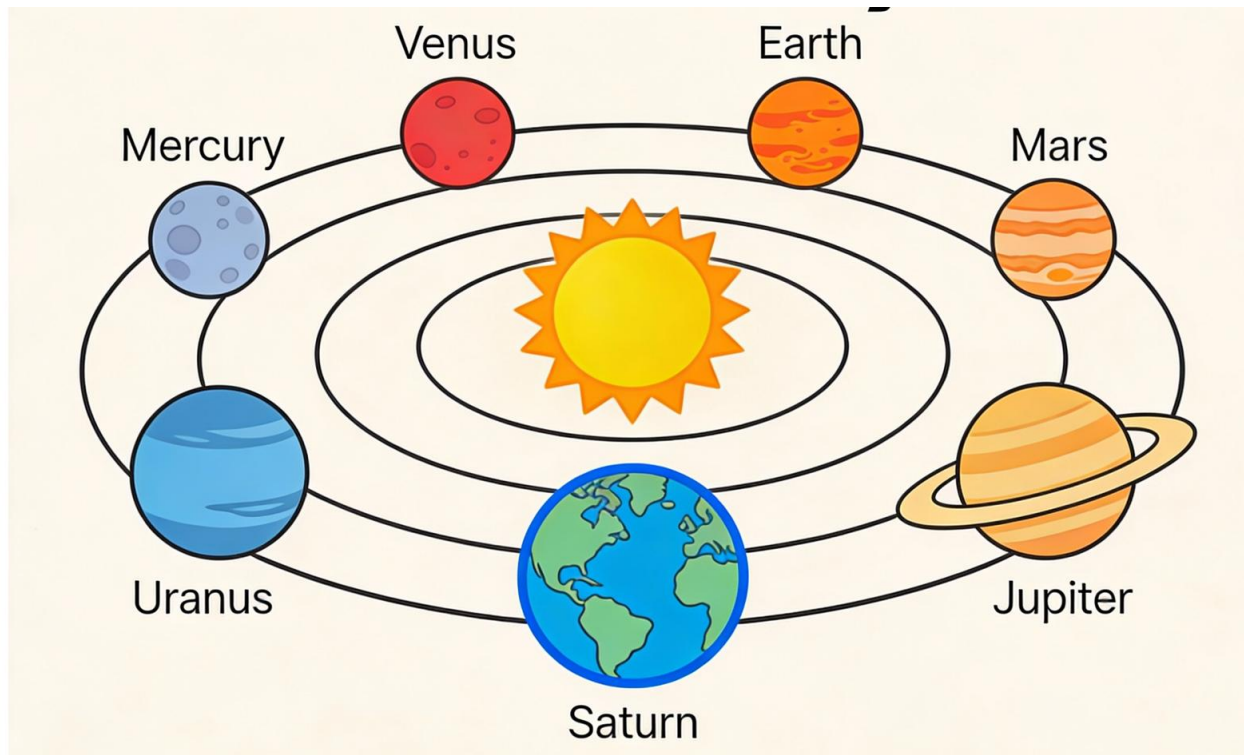
- **The Sun:** It is a huge, very hot star made of burning gases. It is the source of light and heat for the whole solar system. It is NOT a planet.
- **Planets:** These are large, round objects that orbit (go around) the Sun. They do not produce their own light; they reflect the Sun's light. There are **8 planets**.
- **The Moon:** It is a natural satellite that orbits (goes around) our planet, Earth. It reflects the Sun's light, which is why we see it shining at night.
- **Stars:** They are like distant suns, but they are so far away they look like tiny points of light.

The Eight Planets (In order from the Sun)

We remember them with a sentence: **My Very Excellent Mother Just Served Us Noodles.**

- **Mercury** (Closest to Sun, very hot)
- **Venus** (Very bright, often called the Morning/Evening Star)
- **Earth (Our Home Planet!)** Has water and life.
- **Mars** (The Red Planet)

- Jupiter (The Largest Planet)
- Saturn (Has beautiful rings)
- Uranus
- Neptune



How Day and Night Happen

The Earth is shaped like a ball (a sphere). It spins around like a top around an imaginary line called its **axis**. One complete spin takes **24 hours (one day and one night)**.

- **Daytime:** When the part of the Earth you are on spins to face the Sun, it is daytime. The Sun's light shines on you.
- **Night-time:** When your part of the Earth spins away from the Sun, it is night-time. You are in darkness, facing towards space.

Day and Night Rotation Diagram



EVALUATION

1. What is the name of the bright star at the center of our solar system?
2. How many planets go around the Sun?
3. On which planet do human beings live?
4. Does the Moon make its own light? How do we see it?
5. What causes daytime? (Is it because the Sun moves around the Earth, or because the Earth spins?)

ASSIGNMENT

1. **(Drawing)** Draw a simple picture showing the Sun, the Earth, and the Moon. Use arrows to show that the Earth goes around the Sun, and the Moon goes around the Earth.
2. **(Written)** Write down the names of the first three planets, starting from the one closest to the Sun.
3. **(Practical/Observation)** On a clear night, look up at the sky. Can you see the moon? Can you count at least 5 stars? Write "Yes" or "No" for what you saw.

WEEK 10: WEATHER AND CLIMATE

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define weather.
2. List four elements of weather (Temperature, Rainfall, Wind, Sun).
3. Identify weather instruments and what they measure.
4. Describe how to dress for different weather conditions.

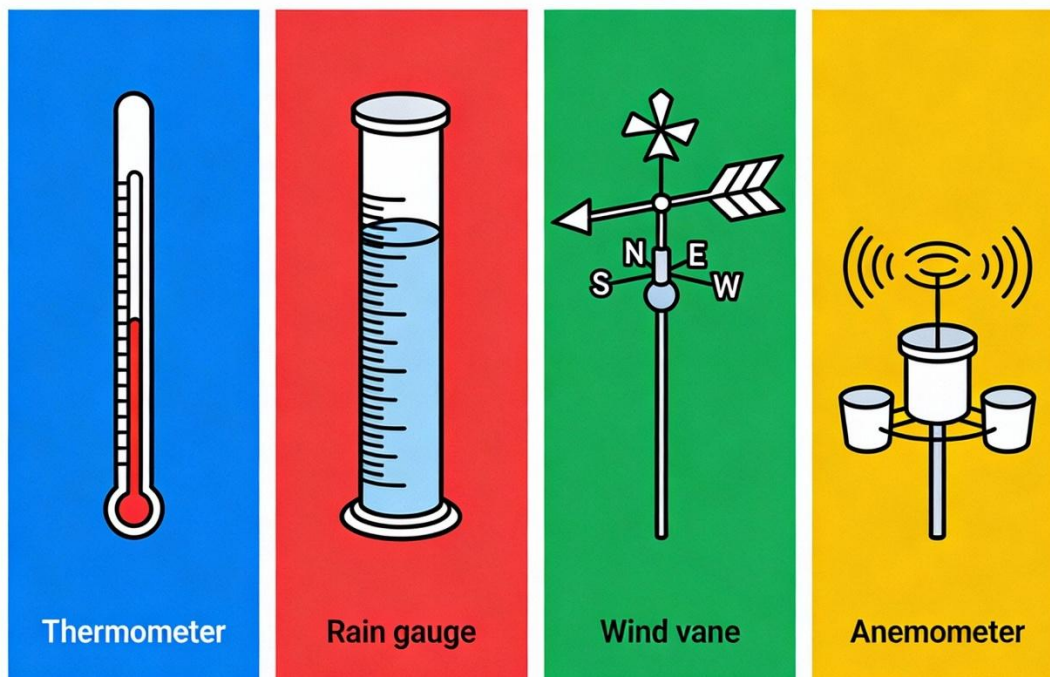
CONTENT

What is Weather?

Weather is the condition of the atmosphere (air around us) at a **particular place and time**. Is it hot, cold, rainy, or windy **right now**? That is the weather. It changes daily.

Elements of Weather (What makes up weather?)

1. **Temperature:** How hot or cold the air is. Measured with a **Thermometer** (°C).
2. **Rainfall:** The amount of rain that falls. Measured with a **Rain Gauge** (in millimeters).
3. **Wind:** Moving air. We measure:
 - **Wind Direction:** Where the wind is blowing FROM. Measured with a **Wind Vane** (points North, South, East, West).
 - **Wind Speed:** How fast the wind is blowing. Measured with an **Anemometer**.
4. **Sunshine/Clouds:** How much of the sky is covered by clouds. We observe this with our eyes.
5. **Humidity:** How much water vapour is in the air. Makes us feel sticky.



Dressing for the Weather

We wear different clothes to stay comfortable and safe in different weather.

- **Hot and Sunny Weather:** Wear **light-coloured, loose cotton clothes** (lets air in). Use a hat, sunglasses, and sunscreen to protect from sunburn.
- **Cold or Harmattan Weather:** Wear **thick, warm clothes** like sweaters, jackets, cardigans, socks, and caps. Layering clothes helps.
- **Rainy Weather:** Wear a **raincoat** or use a strong **umbrella**. Wear **rubber boots** to keep feet dry and avoid stepping in puddles. Avoid silk and clothes that take long to dry.

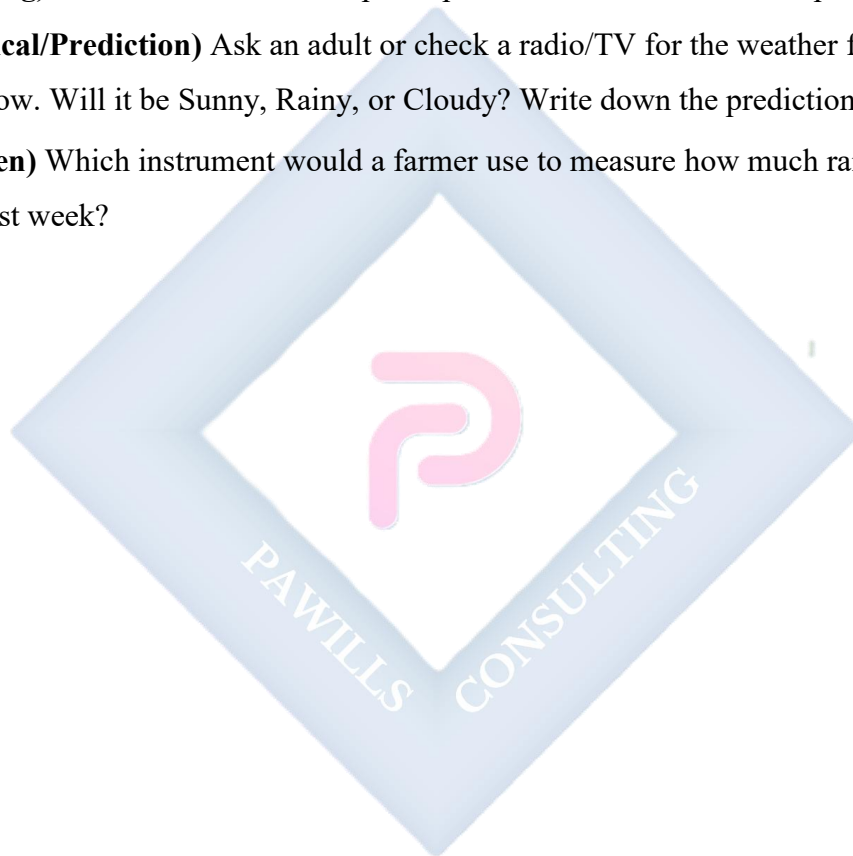
EVALUATION

1. What is the difference between "It is raining today" and "It rains every June in Lagos"? (One is weather, one is climate pattern).

2. If you want to know how hot it is outside, which instrument would you use?
3. What does a wind vane tell us about the wind?
4. What are two things you should wear or carry on a day when the sky is full of dark clouds?
5. Mention two elements that a weather reporter on TV talks about.

ASSIGNMENT

1. **(Drawing)** Draw a wind vane on top of a pole. Label the four cardinal points: N, S, E, W.
2. **(Practical/Prediction)** Ask an adult or check a radio/TV for the weather forecast for tomorrow. Will it be Sunny, Rainy, or Cloudy? Write down the prediction.
3. **(Written)** Which instrument would a farmer use to measure how much rain fell on his farm last week?



WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts from the entire term.
2. Answer questions on Energy, Air, Soil, and Solar System.

CONTENT

Comprehensive Review

- **Energy:** Light (sources, shadows), Electricity (sources, safety), Sound (production, noise), Heat (sources, uses).
- **Environment:** Air (composition, breathing), Soil (types, uses).
- **Space:** Solar System (Sun, planets, Earth), Day and Night.
- **Weather:** Elements, instruments, and dressing appropriately.

EVALUATION

- Oral quiz and written revision exercises covering the term's work.

ASSIGNMENT

- Study your notes from Week 1 to Week 10 for the end-of-term examination.

Here are the detailed and comprehensive lesson notes for **Primary 4 Basic Science & Technology (Third Term)**, strictly following the provided Scheme of Work and the Nigerian (NERDC) Primary Education Curriculum standards.

PRIMARY 4 BASIC SCIENCE & TECHNOLOGY SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Simple Machines	Types of simple machines - lever, pulley, inclined plane, wheel and axle; How simple machines make work easier; Examples in daily life
2	Shape Construction with Paper, Wood or Metal	Basic shapes and their properties; Constructing models with different materials; Tools for construction; Safety in using tools
3	Vehicles	Types of vehicles - land, water, air; Parts of a vehicle; How vehicles move; Transportation in our community; Road safety
4	Technology in Our Daily Lives	Simple technologies at home and school; Communication devices; Importance of technology; Responsible use of technology
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Food and Nutrition	Classes of food; Balanced diet; Food preservation; Healthy eating habits; Local and foreign foods
8	Health and Hygiene	Personal hygiene; Common diseases and prevention; First aid basics; Keeping the environment clean
9	Waste Management	Types of waste; Proper waste disposal; Recycling and reuse; Effects of poor waste management
10	Safety in Our Environment	Safety at home, school and on the road; Identifying hazards; Emergency situations; Helpful safety rules
11	REVISION	Review of all topics covered; Practice exercises; Question and answer sessions
12	EXAMINATION	End of term assessment

13	CLOSING	Report card distribution; Vacation assignment
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PRIMARY 4 BASIC SCIENCE AND TECHNOLOGY LESSON NOTES

THIRD TERM

WEEK 1: SIMPLE MACHINES

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define simple machines.
2. List four types of simple machines (Lever, Pulley, Inclined Plane, Wheel and Axle).
3. Explain how simple machines make work easier.
4. Identify examples of simple machines used in the home.

CONTENT

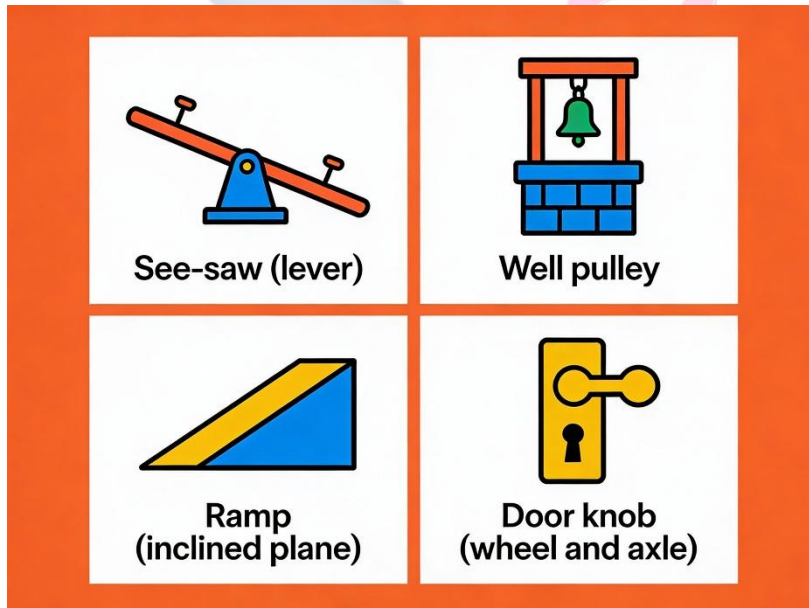
What is a Simple Machine?

A simple machine is a basic tool with few or no moving parts that helps us to do work more easily. "Work" here means moving an object, lifting something, or applying force. Simple machines allow us to use less force, change the direction of force, or increase speed.

Types of Simple Machines

1. **Lever:** A stiff bar that rests on a support called a **fulcrum**. It helps to lift or move heavy loads.
 - *Examples:* See-saw in the park, scissors (two levers joined), bottle opener, broom, crowbar, spade, spoon.
 - *How it helps:* It allows you to lift a heavy rock with a long stick by pushing down on the other end.
2. **Pulley:** A wheel with a grooved rim and a rope or chain passing over it. It is used to lift or lower loads.
 - *Examples:* Flagpole (to raise flag), old well with bucket (draw water), window blinds, crane on a construction site.

- *How it helps:* It lets you lift a heavy bucket of water from a well by pulling down on the rope, which is easier than pulling it straight up.
3. **Inclined Plane (Ramp):** A sloping flat surface that connects a lower level to a higher level.
- *Examples:* A ramp for wheelchairs or goods, staircase, ladder, slide in a playground, road going up a hill.
 - *How it helps:* It is easier to push a heavy box up a ramp than to lift it straight up to the same height.
4. **Wheel and Axle:** Made of a wheel attached to a smaller rod (the axle). When you turn the wheel, the axle turns with more force, or vice-versa.
- *Examples:* Door knob, steering wheel of a car, bicycle wheels, screwdriver, water tap, windmill, pencil sharpener.
 - *How it helps:* A small turn of a big steering wheel makes the car's wheels (attached to the axle) turn a little, giving you control.



How They Make Work Easier They help by:

- Reducing the amount of **force** (effort) we need.
- Changing the **direction** of the force (e.g., pulley pulls down to lift something up).

- Increasing the **speed** or **distance** over which we apply the force.

EVALUATION

1. What do we call tools like levers and pulleys that help us do work easily?
2. Which simple machine is a stiff bar that turns on a fixed point?
3. If you want to move a heavy generator into a truck, would you use a pulley or a ramp?
Why?
4. Is a staircase an example of an inclined plane? What does it help us do?
5. Look around the classroom. Can you name one object that is a lever? (e.g., a pair of scissors on the teacher's table).

ASSIGNMENT

1. **(Drawing)** Draw a pair of scissors and circle the part where the two blades join (the fulcrum of the lever).
2. **(Written)** Look in your kitchen at home. List three different simple machines used there (e.g., bottle opener - lever, knife - wedge, tap - wheel and axle).
3. **(Reasoning)** Why do builders use a pulley to lift cement blocks to the top of a building instead of carrying them up the ladder?

WEEK 2: SHAPE CONSTRUCTION WITH PAPER, WOOD OR METAL

OBJECTIVES

By the end of the lesson, pupils should be able to:

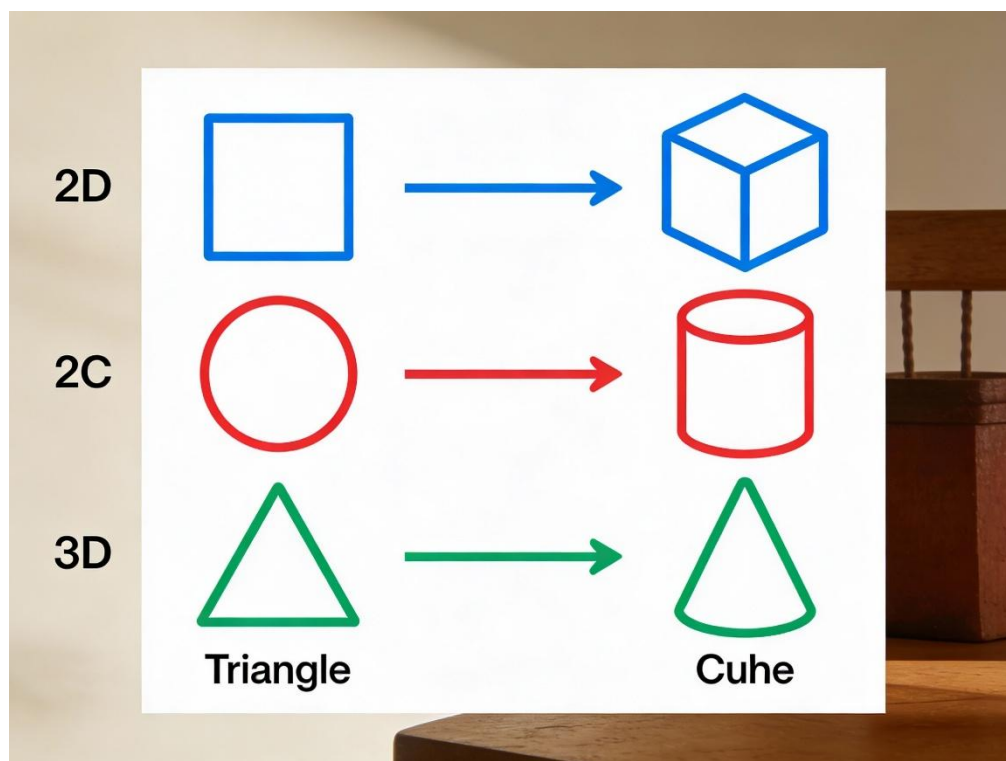
1. Identify basic shapes (Triangle, Square, Rectangle, Cylinder).
2. List materials used for construction (Paper, Wood, Metal).
3. Mention tools used for construction (Scissors, Hammer, Glue).
4. State two safety rules when using tools.

CONTENT

Basic Shapes in Construction

Everything we build is made from basic shapes.

- **Flat Shapes (2D Shapes):** Have length and width.
 - **Square** (all sides equal, 4 corners), **Rectangle** (opposite sides equal, 4 corners), **Triangle** (3 sides, 3 corners), **Circle** (round, no corners).
- **Solid Shapes (3D Shapes):** Have length, width, and height (they are thick).
 - **Cube** (like a sugar cube or a box - made of 6 squares).
 - **Cuboid** (like a book or a brick - made of 6 rectangles).
 - **Cylinder** (like a milk tin, a pipe, or a cup - has two round faces).
 - **Cone** (like a birthday hat or a funnel).



Materials for Construction

We use different materials to make things:

- **Paper/Cardboard:** Light, easy to cut and fold. Used for making cards, paper planes, masks, gift boxes, and models.
- **Wood:** Strong, can be cut and joined. Used for making furniture (tables, chairs), doors, windows, toy cars, and sculptures.
- **Metal:** Very strong and hard, lasts long. Used for making roofs (zinc), pots, gates, cars, bridges, and tools.

Tools for Construction

Each material needs special tools to work with it safely.

Material	Tools Needed	What the tool does
Paper/Card	Scissors, Glue, Ruler, Pencil, Tape	Cuts, sticks, measures, draws.
Wood	Saw, Hammer, Nails, Sandpaper, Chisel	Cuts, joins pieces, smoothens

Material	Tools Needed	What the tool does
		surfaces.
Metal	File, Pliers, Hacksaw, Welding machine, Spanner	Cuts, shapes, joins, tightens.

Safety Rules When Using Tools

1. **Always get permission and supervision** from an adult when using sharp or powerful tools.
2. **Use tools for their correct purpose.** Don't use scissors to cut wire. Don't use a knife as a screwdriver.
3. **Keep your work area tidy** to avoid tripping or losing tools.
4. **Carry sharp tools pointing down** and never run with them.
5. **Wear protection** like goggles if needed (e.g., when hammering).

EVALUATION

1. What shape is a Milo tin? Is it a flat shape or a solid shape?
2. Name two materials that can be used to make a toy car.
3. Which tool would you use to join two pieces of wood together?
4. Why is it dangerous to play with a saw or a pair of scissors?
5. What can you make using cardboard, scissors, and glue?

ASSIGNMENT

1. **(Practical Construction)** Use an empty cardboard box (e.g., milk carton) to make a simple building or a car. Bring it to class.
2. **(Written)** If you visit a carpenter's workshop, name three tools you are likely to see there.
3. **(Drawing)** Draw the shape of a hammer and label the head and the handle.

WEEK 3: VEHICLES

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define a vehicle.
2. Classify vehicles into Land, Water, and Air transport.
3. List the main parts of a vehicle.
4. State two road safety rules.

CONTENT

What is a Vehicle?

A vehicle is any machine that is designed to carry people (passengers) or goods (load) from one place to another.

Types of Vehicles (Based on where they move)

1. **Land Vehicles:** Move on the ground.
 - *On Roads:* Cars, Buses, Motorcycles (Okada), Bicycles, Lorries/Trucks, Keke NAPEP.
 - *On Rails:* Trains.
2. **Water Vehicles:** Move on water (rivers, lakes, seas).
 - *Examples:* Canoes, Boats, Ships, Ferries, Submarines (underwater).
3. **Air Vehicles:** Fly in the air (sky).
 - *Examples:* Aeroplanes, Helicopters, Jets, Hot air balloons, Drones.

Types of Vehicles Chart

Land Vehicles



Car



Truck



Bicycle

Water Vehicles



Boat



Ship



Submarine

Air Vehicles



Plane



Helicopter



Drone

Main Parts of a Land Vehicle (e.g., Car)

- **Body/Chassis:** The main frame and outer shell that holds everything.
- **Engine:** The "heart". It burns fuel (petrol/diesel) to produce power to move.
- **Wheels and Tyres:** Round parts that roll on the ground. Tyres have grooves (treads) to grip the road.
- **Steering Wheel:** Used by the driver to control the direction (left/right).
- **Brakes:** Used to slow down or stop the vehicle.
- **Lights:** Headlights (to see at night), Brake lights (to show you are stopping), Indicators (to show turning).

Road Safety Rules (For Pedestrians - People Walking)

It is very important to be safe on the road.

1. **Cross at Safe Places:** Use a Zebra Crossing, Pedestrian Bridge, or cross where you can see far in both directions.

2. **Stop, Look, Listen:** Before crossing, **STOP** at the curb. **LOOK** left, right, and left again. **LISTEN** for traffic sounds.
3. **Walk, Don't Run:** Walk quickly but do not run across the road.
4. **Walk Facing Traffic:** If there is no pavement (sidewalk), walk on the side of the road facing oncoming vehicles so you can see them.
5. **Be Visible:** Wear bright or reflective clothes when walking at dusk or night.

EVALUATION

1. What is the main work of any vehicle?
2. Give two examples of vehicles that carry many people at once on land.
3. What is the name of a vehicle that moves on water and is often used for fishing in villages?
4. Which part of a car does the driver turn to go left or right?
5. What should you always do before stepping onto a road to cross?

ASSIGNMENT

1. **(Drawing)** Draw a car and label these three parts: Wheels, Window, Headlight.
2. **(Written)** On your way home from school today, take note. Write down the names of three different types of land vehicles you see.
3. **(Reasoning)** Which type of transport (Land, Water, or Air) is the fastest for travelling from Lagos to London? Why?

WEEK 4: TECHNOLOGY IN OUR DAILY LIVES

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define technology simply.
2. Identify simple technologies used at home and school.
3. List three communication devices (Phone, Radio, TV).
4. State the importance of technology.

CONTENT

What is Technology?

Technology is the use of scientific knowledge and skills to create tools, machines, and systems to solve problems and make our lives easier, better, and more fun.

Technology at Home These are gadgets and appliances that use electricity or simple mechanics:

- **In the Kitchen:** Gas or electric cooker, Blender (for smoothies and pepper), Refrigerator (preserves food), Electric kettle.
- **For Cleaning:** Washing machine, Vacuum cleaner, Electric iron.
- **For Comfort:** Fan, Air conditioner, Electric heater.
- **For Information/Entertainment:** Television, Radio, DVD player.

Technology at School

- **For Learning:** Computers, Projectors, Interactive whiteboards, Calculators, Public Address system (microphone).
- **For Administration:** Biometric fingerprint scanners for attendance, printing machines.

Communication Technology

These devices help us send and receive information quickly over long distances.

1. **Mobile Phones/Smartphones:** For calling, texting, taking photos, and accessing the internet.
2. **Radio:** For listening to news, music, and educational programs.
3. **Television (TV):** For watching news, movies, cartoons, and sports.
4. **Computers with Internet:** For sending emails, doing research, video calls (like Zoom), and social media.

Importance of Technology

- **Makes Work Easier and Faster:** A blender grinds pepper in seconds versus minutes with a stone.
- **Improves Communication:** You can talk to a relative in another country instantly.
- **Provides Information and Education:** You can learn anything from the internet.
- **Improves Health:** Medical equipment like X-ray machines help doctors.
- **Provides Entertainment:** TV, video games, music players.

Responsible Use of Technology

- **Balance is key:** Do not spend all day on phones/TV. Play outside, read books, talk to family.
- **Be safe online:** Do not talk to strangers or share personal information on the internet.
- **Handle with care:** Gadgets are expensive and delicate.

EVALUATION

1. In your own words, what is technology?
2. Name one technology tool your mother uses to cook food faster than using firewood.
3. Which device can you use to both listen to music and call your friend?
4. How does a computer help you in school?
5. Why is it important to sometimes put your phone or tablet away?

ASSIGNMENT

1. **(Drawing)** Draw the shape of a mobile phone. Add a smiley face on its screen.

2. **(Written)** Do a quick walk-through of your house. List three electronic gadgets (that use electricity) in your sitting room.
3. **(Reasoning)** Explain how a gas cooker makes cooking easier than using a traditional firewood stove (mention at least one reason).



WEEK 5: MIDTERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

Attempt questions covering topics from Week 1 to Week 4 correctly.

CONTENT

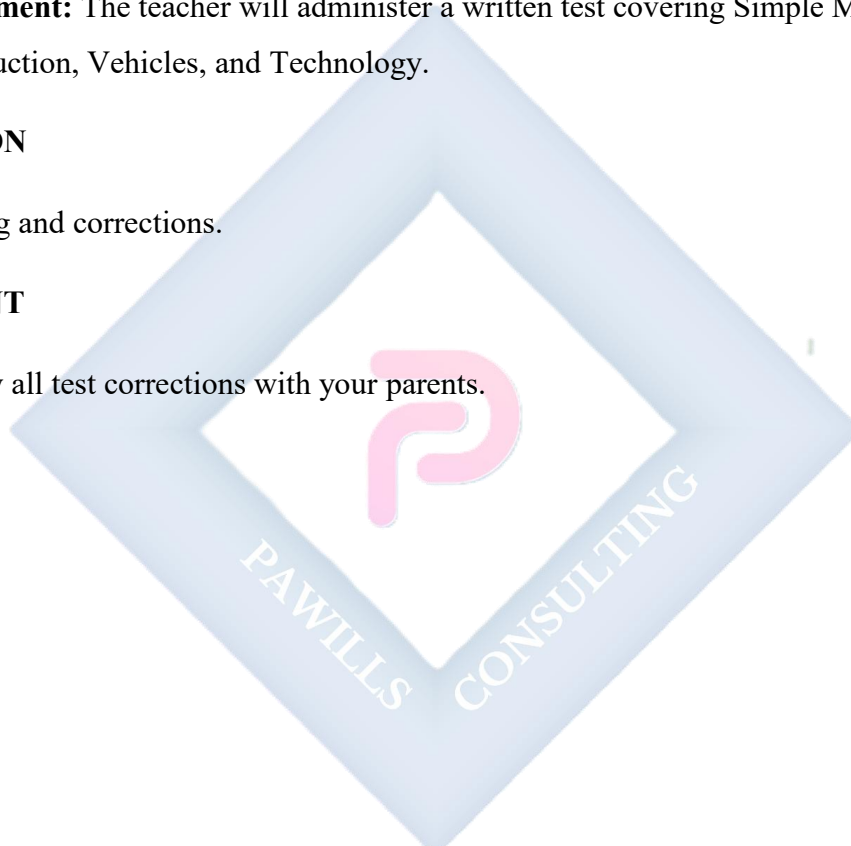
- **Assessment:** The teacher will administer a written test covering Simple Machines, Construction, Vehicles, and Technology.

EVALUATION

- Grading and corrections.

ASSIGNMENT

- Review all test corrections with your parents.



WEEK 6: MIDTERM BREAK



WEEK 7: FOOD AND NUTRITION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define food.
2. List the six classes of food.
3. Explain what a balanced diet is.
4. Differentiate between local and foreign foods.

CONTENT

What is Food?

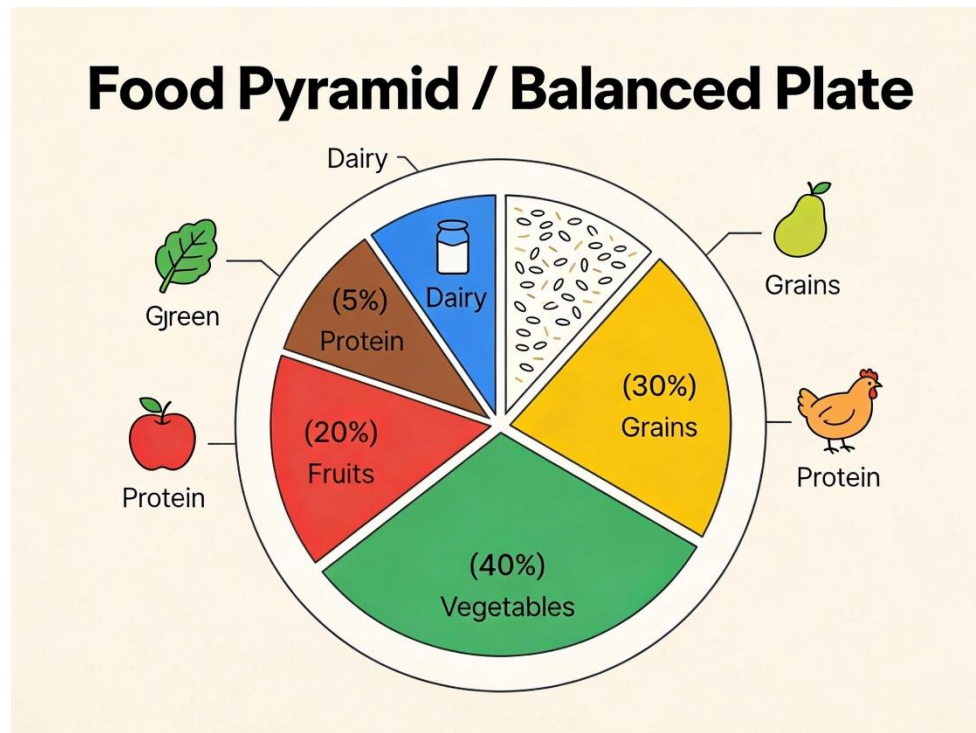
Food is any substance that living things eat or drink to get energy, grow, stay healthy, and repair their bodies.

The Six Classes of Food (Nutrients)

Our body needs different types of food, called nutrients, for different jobs.

1. **Carbohydrates (Energy-giving foods):** They give us energy to work and play.
 - *Examples:* Rice, Yam, Garri, Bread, Maize, Potato, Spaghetti.
2. **Proteins (Body-building foods):** They help to build and repair our muscles, bones, skin, and blood.
 - *Examples:* Beans, Meat, Fish, Egg, Milk, Cheese, Soya beans.
3. **Fats and Oils (Energy storage and warmth):** They give a lot of energy, keep our body warm, and protect organs.
 - *Examples:* Palm oil, Groundnut oil, Butter, Margarine, Avocado.
4. **Vitamins (Protective foods):** They protect our body from diseases and keep our eyes, skin, and gums healthy.
 - *Examples:* Fruits (Orange, Mango, Pawpaw, Watermelon), Vegetables (Spinach, Carrot, Tomato, Pumpkin leaves).

5. **Minerals (Protective and strengthening):** They make our bones and teeth strong and help our blood.
- *Examples:* Iron (in liver and leafy greens), Calcium (in milk, yoghurt, *ukwa*), Iodine (in iodized salt).
6. **Water:** Not a class of food but essential. It helps digestion, carries nutrients, and regulates body temperature. Drink plenty of clean water daily.



Balanced Diet

A balanced diet is a meal that contains the right amounts of different classes of food that your body needs to stay healthy. Eating only *eba* and soup with no fish or vegetables is **not** balanced. A balanced meal could be: Rice + Stew with Fish + Fried Plantain + a slice of Watermelon.

Local Foods vs. Foreign Foods

- **Local/Nigerian Foods:** Foods traditionally grown, prepared, and eaten in Nigeria. They are often more affordable and suited to our climate.
 - *Examples:* Pounded Yam, Amala, Eba, Fufu, Jollof Rice, Moi Moi, Akara, Suya.
- **Foreign Foods:** Foods that originated from other countries but are now eaten in Nigeria.

- *Examples:* Pizza, Burger, French Fries, Shawarma, Spaghetti Bolognese, Chinese rice.

Both can be part of a balanced diet, but local foods are our heritage and usually very nutritious.

EVALUATION

1. Which class of food would an athlete eat a lot of before a race to get energy?
2. If a child needs to grow tall and strong, which class of food should they eat plenty of?
3. What do we call a meal that has carbohydrates, proteins, and vegetables together?
4. Give two examples of Nigerian local foods that you enjoy.
5. Why is it important to eat fruits like oranges?

ASSIGNMENT

1. **(Drawing)** Draw a dinner plate. Divide it and draw small pictures of food to show a balanced meal (include something from carbs, proteins, and vitamins).
2. **(Written)** Write down the names of the six classes of food.
3. **(Reflection)** Think about your lunch yesterday. Did it have food from at least three different classes? List what you ate and its class.

WEEK 8: HEALTH AND HYGIENE

OBJECTIVES

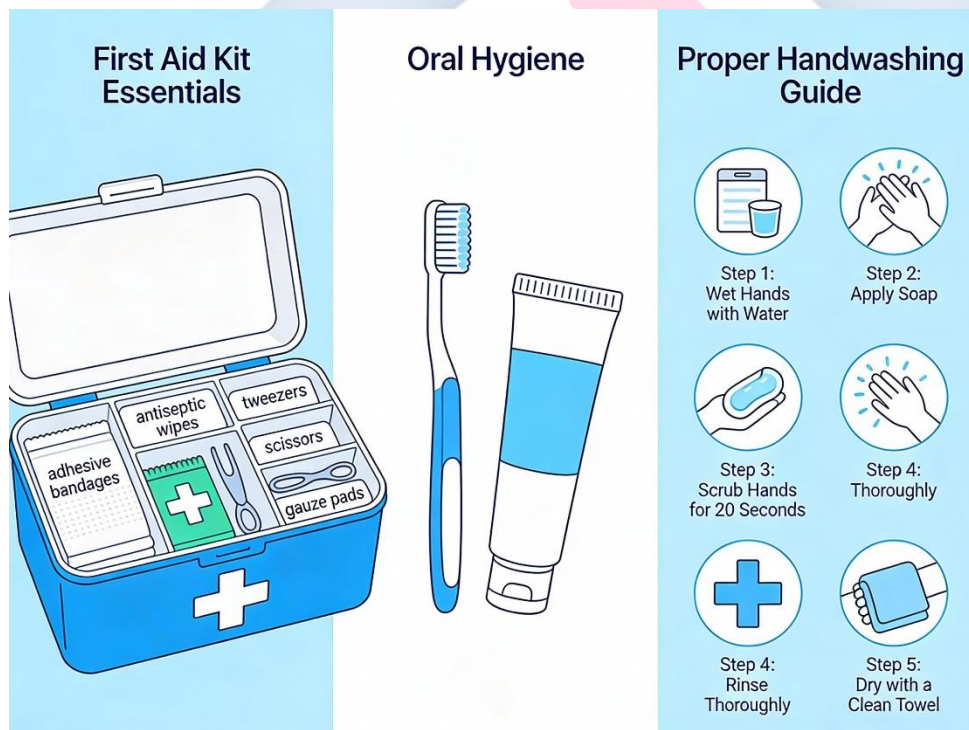
By the end of the lesson, pupils should be able to:

1. Define personal hygiene.
2. List three ways to keep the body clean.
3. Mention two common diseases caused by dirt.
4. State the basics of First Aid.

CONTENT

What is Personal Hygiene?

Personal hygiene means taking care of your own body to keep it clean and healthy. It helps to prevent sickness and makes you look and feel good.



Ways to Keep the Body Clean

1. **Bathing:** Bath at least **once daily**, and twice if you sweat a lot. Use soap, sponge, and clean water to wash off dirt and germs.
2. **Oral Care: Brush your teeth** twice a day (morning and before bed) with toothpaste to prevent tooth decay and bad breath.
3. **Hand Washing:** This is VERY important. Wash hands with soap and running water:
 - **Before** eating or cooking.
 - **After** using the toilet.
 - **After** playing outside or touching animals.
 - **After** coughing or sneezing.
4. **Hair Care:** Wash your hair regularly with shampoo. Comb it daily to keep it neat and free of lice.
5. **Nail Care:** Cut your fingernails and toenails short and keep them clean. Long nails trap dirt and germs.

Common Diseases Caused by Poor Hygiene

- **Cholera & Typhoid:** Caused by drinking dirty water or eating food with germs from dirty hands.
- **Malaria:** Caused by mosquito bites. Mosquitoes breed in dirty, stagnant water around our environment.
- **Ringworm & Scabies:** Skin infections caused by fungi or mites, spread through dirty skin, clothes, or contact.
- **Diarrhea:** Often caused by germs from unwashed hands or contaminated food/water.

First Aid Basics

First Aid is the **first** and immediate help given to a person who is injured or suddenly falls ill, **before** a doctor or nurse arrives.

- **Purpose:** To preserve life, prevent the condition from worsening, and promote recovery.
- **First Aid Box:** A special box containing items to give first aid. It should contain:
 - Cotton wool, Gauze, Bandages, Plasters (Band-Aid), Scissors, Tweezers, Antiseptic (like iodine or hydrogen peroxide), Gloves.

- **Simple First Aid:**

- For a small **cut**: Clean with antiseptic, cover with a plaster.
- For a **nosebleed**: Sit up, tilt head slightly forward, pinch the soft part of the nose.
- **Always call an adult for help** in case of serious injury.

EVALUATION

1. What does "personal hygiene" mean?
2. When are the two most important times to brush your teeth?
3. Name one sickness you can get from eating with dirty hands.
4. How can we prevent mosquitoes from breeding around our house?
5. What is the meaning of "First Aid"?

ASSIGNMENT

1. **(Written)** List four items that should be inside a First Aid box.
2. **(Drawing)** Draw a toothbrush and a tube of toothpaste.
3. **(Practical)** With permission, check your school's First Aid box (or ask your teacher).
Name one item you saw inside it.

WEEK 9: WASTE MANAGEMENT

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define waste.
2. Identify types of waste (solid and liquid).
3. Explain proper waste disposal methods.
4. State the meaning of Reduce, Reuse, and Recycle.

CONTENT

What is Waste?

Waste is any material that is no longer useful or wanted and is thrown away. It is what we call "rubbish" or "refuse".

Types of Waste

1. **Solid Waste:** Dry or wet rubbish that we can pick up.
 - *Examples:* Paper, Plastic bottles/bags, Cans/tins, Food scraps (peels, bones), Broken glass, Old clothes.
2. **Liquid Waste:** Waste in liquid form.
 - *Examples:* Dirty water from bathing, washing clothes/dishes, Urine, Industrial chemicals (from factories).

Proper Waste Disposal Methods

Throwing waste anywhere is dangerous. We must dispose of it properly.

- **Use Dustbins:** Have separate dustbins at home and school. Always use a bin with a lid to keep flies away.
- **Waste Collection:** Government waste trucks (PSP) or private cart-pushers collect waste from bins.
- **Sanitary Landfill:** Waste is buried in a big, specially designed pit far from homes.

- **Incineration:** Burning dry waste in a special high-temperature furnace. **Do not burn plastics openly** as the smoke is poisonous.
- **Composting:** Burying organic waste like yam peels, leaves, and food scraps in a pit. They decay and turn into natural manure for plants.

The 3 Rs of Waste Management (The Best Way!)

1. **Reduce:** Try to use **less** of things that create waste. Say no to extra plastic bags. Use a reusable water bottle instead of buying sachet water every time.
2. **Reuse:** Find a **new use** for an item instead of throwing it away.
 - *Examples:* Use old newspapers to wrap items or clean windows. Use empty plastic bottles as plant pots or pencil holders. Donate old clothes.
3. **Recycle:** Sending waste materials to factories where they are processed and made into **new** products.
 - *Examples:* Old newspapers can be recycled into new paper. Plastic bottles can be recycled into polyester for clothes or new plastic items.



Effects of Poor Waste Disposal

- **Diseases:** Attracts flies, mosquitoes, and rats which spread diseases.

- **Bad Smell (Air Pollution):** Makes the environment stink.
- **Water Pollution:** When waste blocks drains or is thrown into rivers, it causes flooding and poisons water.
- **Soil Pollution:** Chemicals from waste can poison the soil.

EVALUATION

1. What is the difference between solid waste and liquid waste? Give one example of each.
2. Where should you throw a sweet wrapper at school?
3. What is the problem with throwing rubbish into the gutter or drainage?
4. If your plastic water bottle is empty, what is better than throwing it away? (Hint: Think of the 3 Rs)
5. What does "recycle" mean?

ASSIGNMENT

1. **(Drawing)** Draw a picture of a covered dustbin.
2. **(Written)** List three common things you throw away in your house every day.
3. **(Practical/Creative)** Think of one empty container at home (e.g., milk tin, plastic bottle). What can you reuse it for? Write down your idea.

WEEK 10: SAFETY IN OUR ENVIRONMENT

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify hazards (dangers) at home and school.
2. List three safety rules at home.
3. List three safety rules on the road.
4. Explain what to do in an emergency.

CONTENT

Identifying Hazards

A hazard is anything that can cause harm, injury, or an accident. We must learn to spot hazards to stay safe.

- **Hazards at Home:**
 - **In the Kitchen:** Sharp knives left unattended, hot stove, hot oil, slippery wet floor, exposed electrical wires, matches within children's reach.
 - **In the Bathroom:** Slippery floor (soapy water), medicines not locked away.
 - **Living Area:** Broken furniture, unstable shelves, loose electrical cords on the floor.
- **Hazards at School:**
 - **In the Classroom:** Broken chairs/desks, sharp edges on furniture, heavy books falling from shelves.
 - **On the Playground:** Rough play on swings or slides, broken equipment, running on concrete.
 - **General:** Staircases without railings, crowded doorways.

Safety in Our Environment

Road Safety



Yield



Pedestrian Crossing

Home Hazards



Slippery Floor



Sharp Objects



Hot Surface

Emergency Symbols



First Aid



Fire Exit



Evacuation Route

Safety Rules at Home

1. **Kitchen Safety:** Do not play in the kitchen. Turn pot handles inward on the stove. Keep sharp objects in safe drawers.
2. **Electrical Safety:** Do not touch sockets with wet hands. Do not put objects into plug holes. Tell an adult about faulty plugs or wires.
3. **Fire Safety:** Do not play with matches, lighters, or candles. Know how to exit the house in case of fire.
4. **Fall Prevention:** Wipe up spills immediately. Do not climb on high furniture or window sills.

Safety Rules on the Road (For Pedestrians and Passengers)

1. **For Pedestrians:** Use the sidewalk/pavement. Cross at designated points (Zebra crossing, footbridge). Obey traffic lights (Red=Stop, Green=Go).

2. **For Passengers in a Vehicle:** Always wear a seatbelt if available. Do not distract the driver. Keep hands and head inside the vehicle.
3. **General:** Do not play on the road or near parked cars.

What to Do in an Emergency

An emergency is a serious, unexpected situation needing immediate action (e.g., fire, serious injury, someone fainting).

1. **Stay Calm:** Do not panic. Panic makes things worse.
2. **Ensure Your Own Safety First:** Do not run into a burning building. Do not touch a person who is being electrocuted (turn off power first).
3. **Call for Help Immediately:**
 - Shout for an adult nearby.
 - In Nigeria, call the emergency number **112** from any phone. You can also call **767** for Lagos State Emergency.
 - Be ready to tell them your location (address) and what happened.
4. **Give First Aid:** If you know how and it is safe, give basic first aid while waiting for help.

EVALUATION

1. What is a "hazard"? Give one example in your classroom.
2. Why is it dangerous to leave a pot handle sticking out over the edge of a stove?
3. What should you do if you see water spilled on the floor at home?
4. Where is the safest place for you to cross a busy road?
5. What is the universal emergency number you can call in Nigeria for help?

ASSIGNMENT

1. **(Drawing)** Draw a simple "STOP" sign or a "WET FLOOR" caution sign.
2. **(Written)** List two unsafe things you should NEVER do while playing in the school playground.
3. **(Oral/Preparedness)** Memorize your full home address and one parent's phone number. Practice saying it clearly. (This is for telling emergency services).

WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts from the entire term.
2. Answer questions on Machines, Vehicles, Health, and Environment.

Comprehensive Review

- **Simple Machines:** Levers, Pulleys, Ramps, Wheels. How they help us.
- **Construction:** Shapes, materials (paper, wood, metal), tools, and safety.
- **Transport:** Types of vehicles (Land, Water, Air), parts of a car, road safety.
- **Technology:** Definition, examples at home/school, communication devices.
- **Health:** Classes of food, balanced diet, personal hygiene, first aid.
- **Environment:** Waste types, proper disposal, the 3 Rs (Reduce, Reuse, Recycle), identifying hazards, safety rules.

EVALUATION

- Oral quiz and written revision exercises covering the term's work.

ASSIGNMENT

- Study your notes from Week 1 to Week 10 for the end-of-term examination.